

Detection of Invasive Ductal Carcinoma (IDC)

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Detection of Invasive Ductal Carcinoma (IDC)

A project submitted to the

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In

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Bachelor's Degree in Computer Science

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This is to certify that the project titled “**Detection of Invasive Ductal Carcinoma (IDC)**” is the genuine work carried out by **Muhammad Uzair Ishfaq & Khadija Ibrahim**, student of BSCS of Computer Science Department, Lahore Garrison University, Lahore during the academic year 2020-24, in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Science and that the project has not formed the basis for the award previously of any other degree, diploma, fellowship or any other similar title.

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DECLARATION

This is to declare that the project entitled “**Detection of Invasive Ductal Carcinoma (IDC)**” is an original work done by undersigned, in partial fulfilment of the requirements for the degree “Bachelor of Science in Computer Science” at Computer Science Department, Lahore Garrison University, Lahore.

All the analysis, design and system development have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university.

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Date: _____

DEDICATION

I dedicated this work to First of all, ALLAH Almighty who glorified us with the knowledge and bravery to complete this responsibility with elegance. Secondly, our affectionate supportive family whose prayers, advises and continuous support played a major role in achieving this goal. Finally, to the Lahore Garrison University and Especially, the department of Computer Science.

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List of Abbreviation

IDC : Invasive Ductal Carcinoma

BC : Breast cancer

DCIS : Ductal Carcinoma in Situ

PET : Positron emission tomography

MRI : Magnetic resonance imaging

CAD : Computer-aided diagnosis

GPUs : Graphics processing units

E.g : Example

CNNs : Convolutional Neural Networks

LR : Logistic regression

RF : Random forest

K-NN : k-nearest neighbors

SVM : Support vector machines

GNB : Gaussian Naive Bayes

DT : Decision trees

ROC : Receiver operating characteristic

MVC : Model-View-Controller

SOA : Service-Oriented Architecture

DFD : Data Flow Diagram

AI IDC : Artificial Intelligence Invasive Ductal Carcinoma

RBAC : Role Based Access Control

Abstract

Breast cancer affects women globally and ranks as the second leading cause of cancer-related deaths among females. The early diagnosis of breast cancer can significantly lower the mortality rate associated with this disease. Conventional methods for identifying breast cancer include non-invasive imaging modalities like mammography and ultrasonography. These techniques provide valuable information regarding the presence and location of tumors; however, they cannot definitively establish the cancer's malignant potential without further examination. Histopathological evaluation, which involves analyzing tissue samples under a microscope, remains the gold standard for determining the degree of malignancy present within the tumor. While manual assessment of these slide samples may be time-consuming, prone to variability, and liable to human error, advances in technology have led to the development of automated systems that streamline this process while enhancing accuracy and consistency. We proposed a DL-based system for the accurate and efficient detection of Invasive Ductal Carcinoma (IDC) in breast histopathology images. Creating a robust and user-friendly web application with backend and integrated it with Ai model trained on **IDC_Net** that automatically detect that aids medical professionals in diagnosing and treating breast cancer at an early stage. The primary purpose of this project is to address the limitations and challenges associated with manual interpretation of histopathology images, such as inter-observer variability and potential diagnostic errors. By leveraging advanced deep learning methodologies, the proposed system aims to achieve a high level of accuracy in the detection of IDC, thereby facilitating timely and accurate medical interventions for patients. Furthermore, the development of a user-friendly web application interface will ensure seamless accessibility and interpretation of results for healthcare professionals. Ultimately, this project aims to contribute to the advancement of medical technology and improve the standard of care for individuals diagnosed with breast carcinoma, thereby reducing the mortality rate associated with this prevalent form of cancer among women. At the end there no system till now created that can efficiently detect ductal carcinoma pathology images. Our Future direction are simple we are going to implement this system in microscope which will enable us to sell this as product also. Along with that we will continuously enhances its accuracy with due time. Moving toward its mobile application which will make it more easy and handy to work with this system.

INTRODUCTION

Breast cancer (BC) is a widespread malignancy that has emerged as the second leading cause of cancer-related fatalities among women globally over the past few years. Common manifestations of BC include sensations of tightness, soreness, puffiness, or redness in the breast region, along with irregularities like diminished milk production, bloody discharge, or nipple ulcers [1]. Malignant cells typically grow within either the milk ducts or lobules of the breast and produce a lump termed a tumor. Based on the location and nature of the cancerous growth, the tumor can be classified into two main categories: Ductal Carcinoma in Situ (DCIS) and Invasive Ductal Carcinoma (IDC). IDC progressively spreads to distant regions of the body, whereas DCIS remains confined to the original site. Unfortunately, invasive cancer accounts for approximately 80% of BC-related mortality, highlighting its deadly potential. Moreover, BC affects younger populations as well, particularly those in their thirties and forties, with an alarming increase in incidence rates globally. According to international cancer registries and the Global Initiative for Cancer Registry Development, the global number of newly diagnosed BC cases and deaths in both men and women was estimated at 2,261,419 and 684,996, respectively, in 2022 [2].

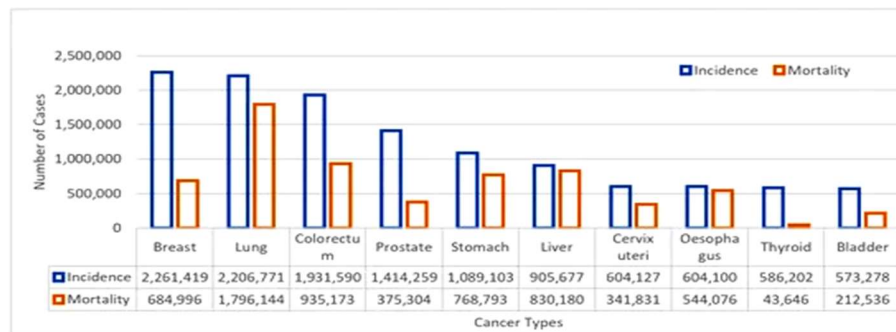


Figure 1: Estimated number of incident cancer cases and deaths worldwide due to BC

In the diagnostic process for Breast Cancer (BC), multiple screening techniques are employed, including thermography, mammography, ultrasound scans, positron emission tomography (PET), and magnetic resonance imaging (MRI). These modalities provide valuable information regarding the location and size of any tumors present within the breast tissue. Moreover, microscopic

examination through biopsy or surgical excision is considered the gold standard for accurately determining the nature and extent of the disease. Nonetheless, the classification of BC subtypes according to histopathological features such as tissue texture, coloration, and shape distinctions can be problematic due to the subtle differences in these characteristics between distinct subtypes. Specifically biopsy play a vital role in early detection Pathologists typically identify this type by visually examining tissue samples under a microscope but limitations and challenges associated with manual interpretation of histopathology images, such as inter-observer variability and potential diagnostic errors. That leads to wrong diagnose and can lead to loss of a life. Researchers are developing computer-aided diagnosis (CAD) systems based on artificial intelligence techniques to overcome such challenges. Deep neural networks have gained widespread adoption in the development of computer-aided design (CAD) systems owing to their inherent capabilities for automated feature extraction, multi-layered representations, and learning abilities. These advanced features have been facilitated by significant advancements in computer vision and the increasing accessibility of computing resources, particularly graphics processing units (GPUs). The extensive training process necessitates large datasets, additional time, and specialized hardware infrastructure to optimize the performance of these complex models. In an effort to enhance the current standards of breast invasive ductal carcinoma (IDC) classification, our FYP aimed to investigate and optimize the use of histopathological images as a predictive tool through rigorous analysis and comparative evaluation with existing methodologies. By leveraging cutting-edge techniques and advanced computational methods, we sought to improve upon established approaches and establish novel strategies that could facilitate more accurate diagnoses and ultimately contribute to improved patient outcomes. The original dataset comprised 162 whole mount slides of breast cancer (BC) specimens obtained through imaging at 40x magnification. Subsequently, we extracted a total of 277,524 patches with a fixed size of 50x50 pixels from these slides. These patches were categorized based on their clinical classification, resulting in two distinct groups: 198,738 patches belonging to IDC-negative cases and 78,786 patches corresponding to IDC-positive cases. Notably, each patch's filename follows a specific convention consisting of the patient identification number followed by the coordinates of the patch location within the image, separated by underscores, along with its corresponding class label (e.g., "10253_idx5_x1351_y1101_class0" represents a patch located at pixel coordinate (1351, 1101) of the image associated with patient identifier "10253," which belongs to the non-IDC class) [3].

LITERATURE REVIEW

The literature review conducted for the "**Detection of invasive ductal carcinoma (IDC)**" application involves an analysis of several existing research papers. In June 2022 a research paper published by Érika G de Assis , Zenilton K G do Patrocínio , Cristiane Neri Nobre named "The Use of Convolutional Neural Networks in the Prediction of Invasive Ductal Carcinoma in Histological Images of Breast Cancer" used 3-hidden-layer CNN with data balancing reached both accuracy and F1-Score of 0.85 [4]. In another research published on Nov 2022 by Wingates Voon, Yan Chai Hum, Yee Kai Tee, Wun-She Yap, Maheza Irna Mohamad Salim, Tian Swee Tan, Hamam Mokayed & Khin Wee Lai named "Performance analysis of seven Convolutional Neural Networks (CNNs) with transfer learning for Invasive Ductal Carcinoma (IDC) grading in breast histopathological images" used seven selected CNN models: EfficientNetB0, EfficientNetV2B0, EfficientNetV2B0-21k, ResNetV1-50, ResNetV2-50, MobileNetV1, and MobileNetV2 with transfer learning. According to our investigations, the state-of-the-art EfficientNetV2B0-21k model demonstrated exceptional performance in the (IDC) task, with an impressive 0.72 billion floating-point operations and 7.1 million parameters. Notably, despite this advanced architecture, nearly all CNN models evaluated in our study exhibited remarkable accuracy in the IDC grading task, with an average balanced accuracy across the cross-validation and test sets of 0.936 ± 0.0189 and 0.9308 ± 0.0211 , respectively. These results underscore the robustness and generalizability of deep learning approaches for this task [5]. In February 2022 a research paper named "Predicting invasive ductal carcinoma tissues in whole slide images of breast Cancer by using convolutional neural network model and multiple classifiers" by B. G. Deepa & S. Senthil. In this research, an innovative deep learning approach was employed utilizing a sequential convolutional neural network (CNN) architecture consisting of convolutional layers (conv2D), max pooling layers (MaxPooling2D), dropout regularization, flattening, and dense layers to tackle the challenging task of intra-day credit risk assessment (IDC). To compare the performance of the proposed CNN-based framework against established classification methods, a comprehensive evaluation was conducted involving logistic regression (LR), random forest (RF), k-nearest neighbors (K-NN), support vector machines (SVM), linear SVM, Gaussian Naive Bayes

(GNB), and decision trees (DT). These models were trained and evaluated using the IDC dataset, which consists of various features related to financial transactions. The results of the analysis revealed that the proposed CNN-based framework exhibited remarkable performance, achieving an accuracy ranging from 80% to 86%, precision ranging from 92% to 94%, recall ranging from 91% to 96%, and F1-score ranging from 94% to 96%. Moreover, when applied to a separate test dataset, the CNN model demonstrated impressive predictive capabilities, yielding an accuracy of 91% to 94%, precision of 91% to 95%, recall of 93% to 96%, and F1-score of 95% to 98%. These findings suggest that the proposed CNN-based IDC framework offers a highly effective solution for accurately forecasting credit risk. To further elucidate the strengths of the proposed methodology, they investigated the contributions of individual components within the CNN architecture. Their analyses indicated that the convolutional layers and pooling layers played crucial roles in extracting relevant features from the input data, while the dropout regularization helped prevent overfitting and improve generalization performance. Additionally, the use of dense layers enabled the model to learn complex patterns within the data, leading to improved predictions [6].

In one of the latest research publish in june 2023 by Eelandula Kumaraswamy, Sumit Kumar and Manoj Sharma. In this research, they investigate the possibility of utilizing pre-trained convolutional neural networks (CNNs), specifically EfficientNetV2L, ResNet152V2, and DenseNet201, for various tasks in deep learning. These pre-trained models have shown great success in image classification tasks, but their potential applications extend beyond that field. By leveraging these pre-trained models as a starting point, we can accelerate the training process and improve performance on other tasks such as object detection, segmentation, and generation. According to the research, an ensemble model consisting of convolutional neural networks (CNNs) demonstrated impressive performance in classifying images into three grades with high accuracy. Specifically, the model achieved a remarkable 94% overall accuracy across all grade levels (grades 1, 2, and 3), as indicated by the area under the receiver operating characteristic (ROC) curve. The results suggest that the ensemble model outperformed expectations, with notable areas of excellence in grading images belonging to grades 1, 2, and 3, where it recorded accuracies of 96%, 94%, and 96%, respectively [7]. Here is the table representation of research paper of which we have done analysis and then moved to solution based on our experiences thinking which leads to a remarkable results. Table is given below.

Table 1: Table of research paper studied.

NO	Research Paper	Date Of Publication	Methods	Results
1	The Use of Convolutional Neural Networks in the Prediction of Invasive Ductal Carcinoma in Histological Images of Breast Cancer	6-June-2022	CNN	Accuracy and F1-Score of 0.85.
2	Performance analysis of seven Convolutional Neural Networks (CNNs) with transfer learning for Invasive Ductal Carcinoma (IDC) grading in breast histopathological images	10-November-2022	EfficientNetB0, EfficientNetV2B0, EfficientNetV2B0-21k, ResNetV1-50, ResNetV2-50, MobileNetV1, and MobileNetV2 with transfer learning.	average balanced accuracy of 0.936 ± 0.0189 on the cross-validation set and 0.9308 ± 0.0211
3	Predicting invasive ductal carcinoma tissues in whole slide images of breast Cancer by using convolutional neural network model and multiple classifiers	03-January-2023	logistic regression (LR), random forest (RF), k-nearest neighbor (K-NN), support vector machine (SVM), linear SVM, gaussian naïve bayesian (GNB) and decision tree (DT)	80%–86% of accuracy

4	Predicting Invasive Ductal Carcinoma by Using Deep Convolutional Neural Network	27-December-2022	CNN	91.28% accuracy and 0.22 loss.
5	An Invasive Ductal Carcinomas Breast Cancer Grade Classification Using an Ensemble of Convolutional Neural Networks	5-June-2023	EfficientNetV2L, ResNet152V2, DenseNet201	96%, 94%, and 96% Accuracy.

PROBLEM DEFINITION

In consideration of historical instances worldwide wherein erroneous pathology diagnoses resulted in incorrect patient treatment, culminating in misguided pharmaceutical interventions and, ultimately, fatalities, it becomes imperative to mitigate such occurrences. The imperative lies in refining and enhancing the precision of Invasive Ductal Carcinoma (IDC) analyses, not only for the benefit of pathologists but, more importantly, to safeguard the lives of countless individuals. This undertaking seeks to address inherent limitations and challenges inherent in the manual interpretation of histopathology images, including the potential for inter-observer variability and diagnostic errors. Through the strategic application of advanced deep learning methodologies, the proposed system endeavors to attain a heightened level of accuracy in detecting IDC. This pursuit is consequential in facilitating timely and precise medical interventions, thereby significantly improving patient outcomes. Additionally, the development of a user-friendly web application interface is pivotal in ensuring the seamless accessibility and interpretation of results for healthcare professionals. By amalgamating technological innovation with medical diagnostics, this project aspires to contribute substantively to the progress of medical technology. More specifically, it aims to elevate the standard of care provided to individuals afflicted with breast carcinoma, consequently diminishing the mortality rate associated with this pervasive form of cancer among women. In essence, this initiative aligns with the broader objective of advancing medical knowledge and fortifying the healthcare infrastructure to address pressing challenges in the realm of oncology.

Chapter 4

SOFTWARE REQUIREMENT SPECIFICATION

Project involves a comprehensive system for pathology image analysis and management. Let's break down the product perspective with deep details:

1. IDC_Net Model:

- Architecture: Self-made IDC_Net model.
- Training Data: Histopathology images.
- Accuracy: 98.3%.
- File Format: H5 file for the trained model.

2. Web Application (Laravel):

- User Roles: Admin, Laboratory Technician, Pathologist.
- Authentication: Username, Password, Role.
- Admin Dashboard:
 - Counts of Admins, Technicians, Pathologists.
 - Total Positive and Negative Cancer Cases.
 - Total Cases Reported.
 - Treated and Not Treated Cases.
- "Add Logins" View:
 - Add new logins with details (Name, Username, Email, Password, Role, Contact No, and Profile Image).
- "All Logins" View:
 - Display and manage all logins (Edit, Update, Delete).

3. Laboratory Technician Dashboard:

- Counts of Patients added today.
- Last patient treated today.
- Number of patients to be treated.
- "Add Patient" View:
 - Add patient details (Sample No, Name, Email, and Contact No).

- Send email to the patient with Sample No and other details.
- "All Patients" View:
- Display all patient data.
- Edit patient data (Name, Email, and Contact No).
- Send email to the patient on data update.

4. Pathologist Dashboard:

- Counts of total patients today.
- Counts of treated and untreated patients.
- "Appointment" View:
- List of untreated patients.
- "Artificial Intelligence IDC" button for image analysis.
- IDC Prediction View (Flask):
- Choose images for analysis.
- Live detection using the camera.
- Save positive images.
- Update record with positive/negative results.
- Send result emails to patients.

5. Communication:

- Email Notifications:
- Patients receive notifications on data updates and results.
- Internal Communication:
- Pathologists and Technicians can communicate through the system.

6. Security:

- User authentication and authorization.
- Unique usernames.
- Secure password storage.
- Data encryption during transmission.

7. Integration:

- Laravel and Flask integration for seamless user experience.
- Model integration for image analysis.

8. User Experience:

- Intuitive dashboards for quick information access.
- Modal pop-ups for data updates.
- Responsive design for varied devices.

9. Scalability:

- System should handle a growing number of users, logins, and patients.
- Optimized database structure for performance.

10. Maintenance:

- Regular updates for security and feature enhancements.
- Monitoring system health and performance.

11. Documentation:

- Detailed documentation for installation, usage, and troubleshooting.

12. Testing:

- Rigorous testing of all features.
- Unit testing, integration testing, and user acceptance testing.

Type of people who will use the system:

Certainly! Here's a detailed overview of the roles and functionalities for each type of user in system:

1. Admin:

- **Responsibilities:**

- 1. User Management:** The Admin is responsible for managing all users in the system, including Admins, Laboratory Technicians, and Pathologists.
- 2. Dashboard Overview:** Upon logging in, the Admin is presented with a dashboard that provides key statistics such as the counts of different user roles, total positive and negative cancer cases, treated and untreated cases, and the overall count of reported cases.
- 3. User Creation and Management:** The Admin has the authority to add new users by providing essential details such as Name, Username, Email Address, Password, Role (Admin, Laboratory Technician, and Pathologist), Contact Number, and Profile Image.

4. **Viewing and Modifying User Data:** The Admin can view a comprehensive list of all users and has the capability to update user information, ensuring that the system's user database is accurate and up-to-date. This includes the ability to delete user records if needed.
- **Views:**
 1. **Admin Dashboard:** Presents an overview of system statistics.
 2. **Add Logins:** Allows the Admin to add new users with specified details.
 3. **All Logins:** Displays a table of all users, enabling the Admin to view, update, and delete user records.

2. Laboratory Technician:

- **Responsibilities:**
 1. **Patient Management:** The Laboratory Technician is responsible for managing patient data.
 2. **Dashboard Overview:** The Technician's dashboard includes counts of patients added on the current day, the last patient treated, and the number of patients yet to be treated.
 3. **Adding Patient Data:** The Technician has access to the "Add Patient" view where they can input patient details such as Sample Number, Name, Email Address, and Contact Number.
 4. **Email Notification:** Upon adding a patient, an automatic email is sent to the patient, providing them with their Sample Number and other relevant details.
 5. **Viewing and Updating Patient Data:** The "All Patient" view allows the Technician to see a table of all patients, edit patient information, and update records. Patients receive email notifications when their data is updated.
- **Views:**
 1. **Laboratory Technician Dashboard:** Displays an overview of patient-related statistics.
 2. **Add Patient:** Enables the Technician to add new patient data.
 3. **All Patient:** Presents a table of all patients for viewing and updating.

3. Pathologist:

- **Responsibilities:**

1. **Diagnostic Analysis:** The Pathologist is responsible for analyzing diagnostic data, making treatment decisions, and updating patient records.
2. **Dashboard Overview:** The Pathologist's dashboard includes counts of total patients for the day and the number of treated patients or patients whose results have been added.
3. **Appointment View:** Lists patients not yet treated by the pathologist, with an option to perform artificial intelligence-based IDC analysis.
4. **IDC Analysis:** In the "Appointment" view, the Pathologist can choose and analyze images for IDC using a Flask-based interface. The results of the analysis are then updated in the patient's record.
5. **Result Update:** The Pathologist can update patient records with diagnostic results (positive or negative IDC), triggering an automatic email notification to the patient.

- **Views:**

1. **Pathologist Dashboard:** Displays an overview of diagnostic-related statistics.
2. **Appointment:** Lists patients for analysis and IDC detection.
3. **IDC Analysis in Flask:** Provides tools for analyzing pathology images and updating patient records.

METHODOLOGY

Writing a methodology typically involves explaining the step-by-step process of how you conducted your study, experiment, or project. Below is a detailed methodology for the code and web application. The methodology is broken down into sections, and explanations are provided for each step.

1. Data Collection and Preprocessing

- **Data Collection:**

The breast histopathology images dataset was obtained from Kaggle using the Kaggle API. The dataset contains images of breast cancer biopsies, with each image labeled as IDC (Invasive Ductal Carcinoma) positive or negative.

- **Data Preprocessing:**

The images were preprocessed using the following steps: Images were resized to 50x50 pixels. IDC-negative and IDC-positive images were separated. To balance the classes, a subset of IDC-negative images was selected. Images were read using OpenCV, resized, and stored along with their corresponding labels (0 for IDC-negative, 1 for IDC-positive). The dataset was shuffled to ensure randomness.

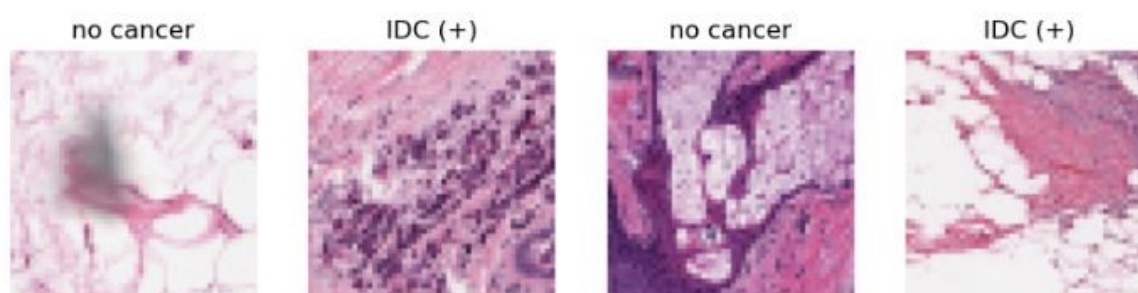


Figure 2: Breast Histopathology Images

2. Model Development

- **Model Architecture:**

An IDC_Net model was implemented using TensorFlow and Keras. The model architecture consists of convolutional layers with batch normalization, max-pooling layers, dropout layers, and dense layers with ReLU activation. The output layer uses a sigmoid activation for binary classification.

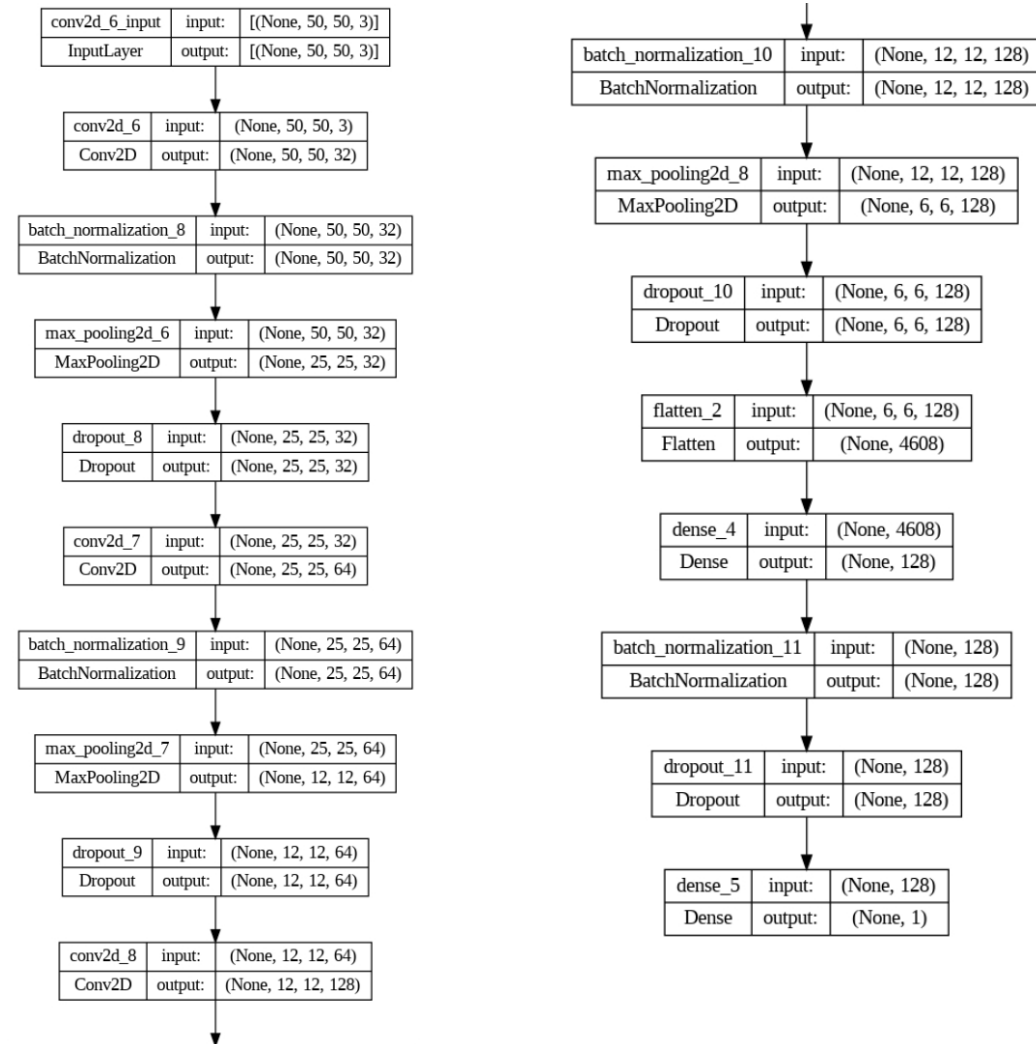


Figure 3: IDC_Net Architecture diagram

- **Data Augmentation:**

ImageDataGenerator was used for data augmentation during training. Augmentation techniques include rotation, width shift, height shift, shear, zoom, and horizontal/vertical flipping.

- **Model Training:**

The model was compiled using the Adam optimizer with a learning rate of 0.0001 and binary cross-entropy loss. Training was performed for 200 epochs with a batch size of 35.

- **Model Evaluation:**

The model was evaluated on the test set, and accuracy and loss were plotted over epochs. A confusion matrix and a heatmap representing the percentage confusion matrix were generated to assess classification performance.

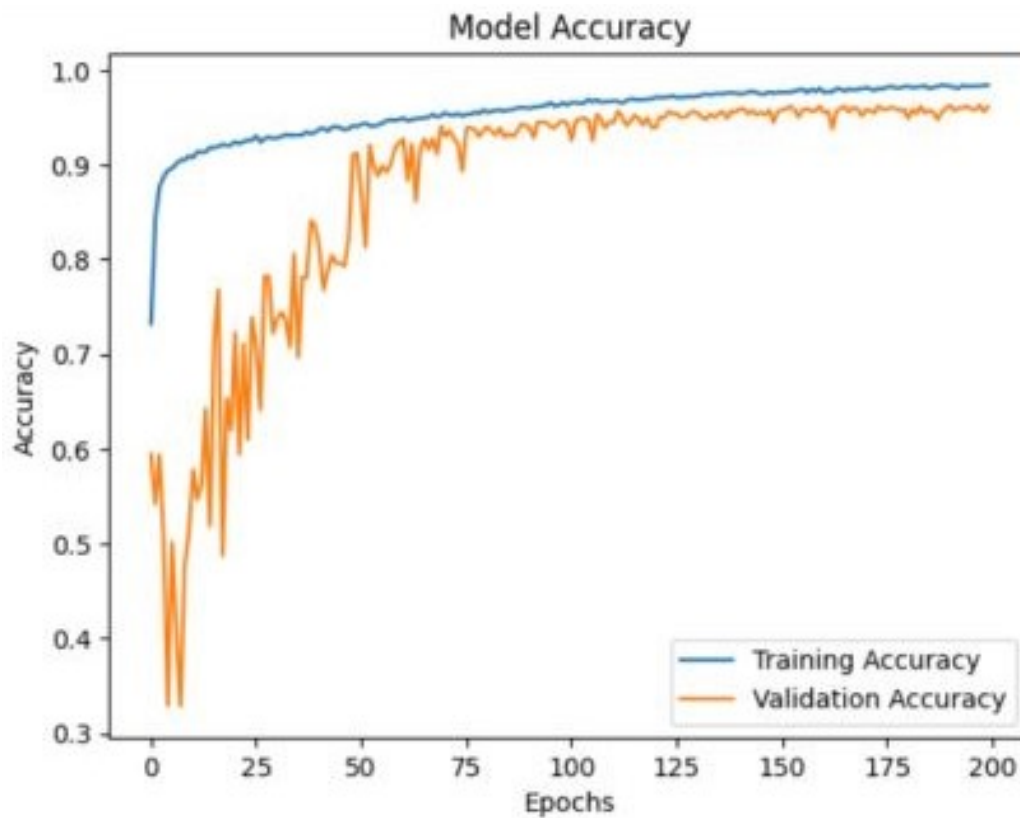


Figure 4: Model Accuracy

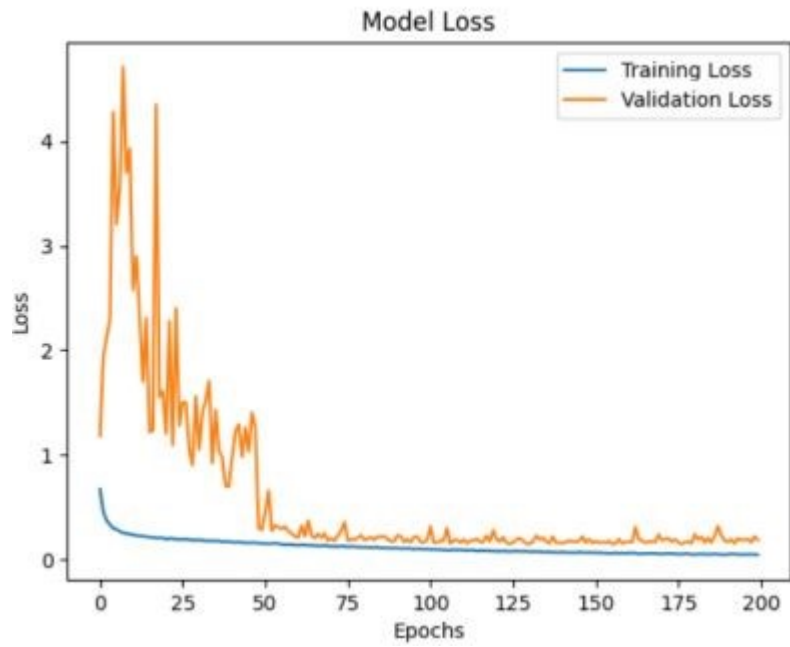


Figure 5: Model Loss

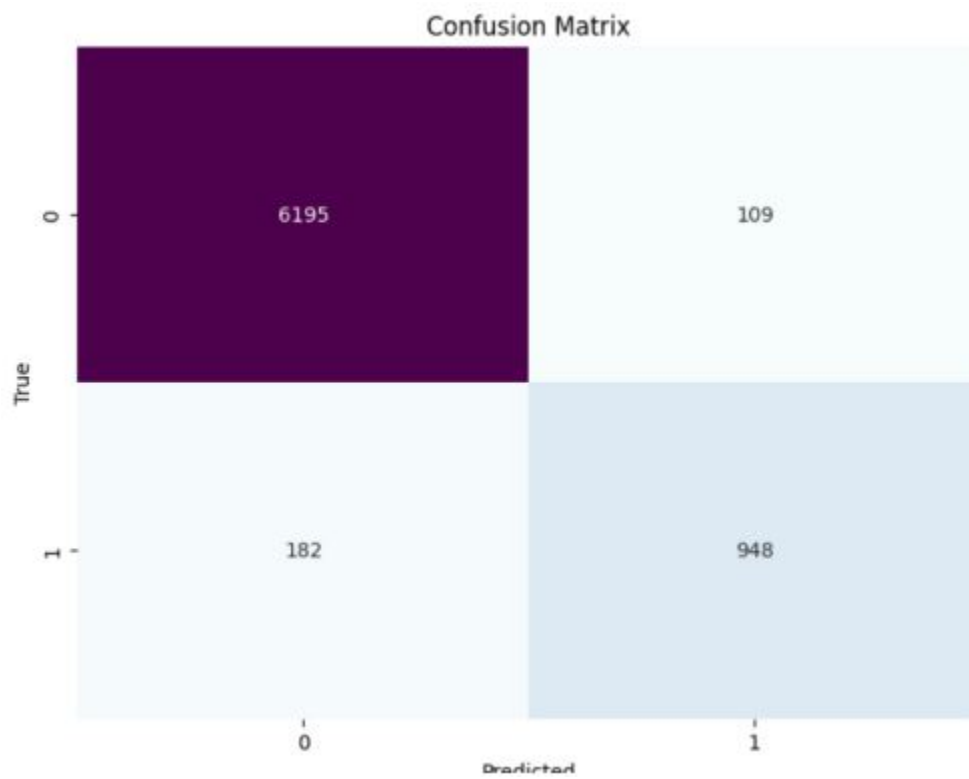


Figure 6: Model Confusion Matrix

- **Model Saving:**

The trained model was saved as "Breast_idc_Model.h5" for future use.

3. Web Application Development

- **Technology Stack:**

The web application was developed using the Laravel framework for the backend and PHP, Flask HTML, CSS, and JavaScript for the frontend.

- **User Authentication:**

The application provides user authentication with different roles: Admin, Laboratory Technician, and Pathologist. Each user type has specific access permissions.

A user authentication form titled "IDC" in red. It contains three input fields: "Username", "Password", and "Select Role" (a dropdown menu). Below these fields is a pink "LOG IN" button.

Figure 7: User Authentication

- **Admin Dashboard:**

The admin dashboard displays counts of Admins, Technicians, Pathologists, total positive and negative cancer cases, total reported cases, treated cases, and pending cases.

- **Admin Functionalities:**

Add Logins: Admin can add logins by providing name, username, email, password, role, contact number, and profile image.

All Logins: Admin can view, update, and delete records of all users.

- **Technician Dashboard:**

The technician dashboard shows counts of patients added today, last patients treated today, and the number of patients to be treated.

- **Technician Functionalities:**

Add Patient: Technician can add patient data, and a confirmation email is sent to the patient.

All Patients: Technician can view, edit (name, email, contact number), and update patient records.

- **Pathologist Dashboard:**

The pathologist dashboard displays counts of total patients today and the number of patients with results pending.

- **Pathologist Functionalities:**

Appointment: Shows a table of patients awaiting treatment. The pathologist can click on the "Artificial Intelligence IDC" button to access the Flask-based image prediction view.

Image Prediction View: Allows pathologist to select biopsy images, perform live detection, and update patient records with results.

4. Flask-Based Image Prediction View

- **Functionality:**

This view provides a selection for uploading biopsy images. The pathologist can select and submit images, and the model predicts IDC-positive or IDC-negative patches.

- **Live Detection:**

A live detection option is available, where the pathologist can use a camera to determine IDC status.

- **Result Selection and Record Update:**

The pathologist can select the result (positive/negative) for each patch, and the overall result is used to update the patient's record. A confirmation email is sent to the patient with the result.

5. Conclusion

This methodology details the entire process, from data collection and model development to web application design and functionality. Each step contributes to the seamless integration of AI-driven cancer diagnosis into a web-based healthcare system.

6. Other model Tried and their accuracy:

Table 2: Table of Accuracy and Loss of model tried

S No	Model	Accuracy	Loss
1	VGG 19	90.4%	24.3%
2	Resnet50	89.9%	25.9%
3	Efficient Net B0	90.6%	23.2%
4	IDC_Net	98.3%	4.3%

DETAILED DESIGN AND ARCHITECTURE

6.1 System Architecture

Overview

The system architecture is designed to seamlessly integrate a breast cancer diagnosis model, developed using histopathology images, into a web application built with the Laravel framework. The architecture encompasses three major components: the Machine Learning Model, the Laravel Web Application, and the Flask-Based Image Prediction Service. Each component plays a distinct role in achieving the overall functionality of the system.

1. Deep Learning Model Component

- **Responsibilities:**

Train and evaluate a breast cancer diagnosis model using histopathology images. Save the trained model for later use in the prediction service.

- **Rationale:**

Separating the machine learning model as a distinct component allows for modularity and easy updates without affecting the web application or prediction service. The model can be retrained independently without impacting the rest of the system.

2. Laravel Web Application Component

- **Responsibilities:**

User authentication and authorization for Admins, Laboratory Technicians, and Pathologists. Admin dashboard with statistical insights and functionalities. Technician dashboard with patient-related functionalities. Pathologist dashboard with patient appointments and access to the image prediction service. Manage user logins, patient records, and system-wide settings.

- **Rationale:**

Laravel provides a robust framework for building web applications with a user-friendly interface. Division of responsibilities allows for role-specific features and a clear separation of concerns. Easy scalability and maintainability of the application.

3. Flask-Based Image Prediction Service Component

- **Responsibilities:**

Accept biopsy images from the Pathologist via the web application. Utilize the pre-trained machine learning model to predict IDC-positive or IDC-negative patches in the images. Provide a live detection option using a camera.

- **Rationale:**

Flask is lightweight and well-suited for creating RESTful services. Separation of image prediction logic from the main web application ensures efficient processing. Flexibility to incorporate future updates or alternative machine learning models without affecting the rest of the system.

4. Collaboration

The Laravel Web Application interacts with both the Machine Learning Model and the Flask-Based Image Prediction Service. Admins, Technicians, and Pathologists use the web application to perform their respective tasks. The Flask service communicates with the web application to receive biopsy images and sends the predictions back.

5. Design Patterns

Model-View-Controller (MVC): The Laravel Web Application follows the MVC pattern, separating concerns for better organization and maintainability.

Service-Oriented Architecture (SOA): The Flask-Based Image Prediction Service operates independently as a service, promoting modularity and scalability.

6. Diagram

A high-level diagram illustrates the interactions between the three components:

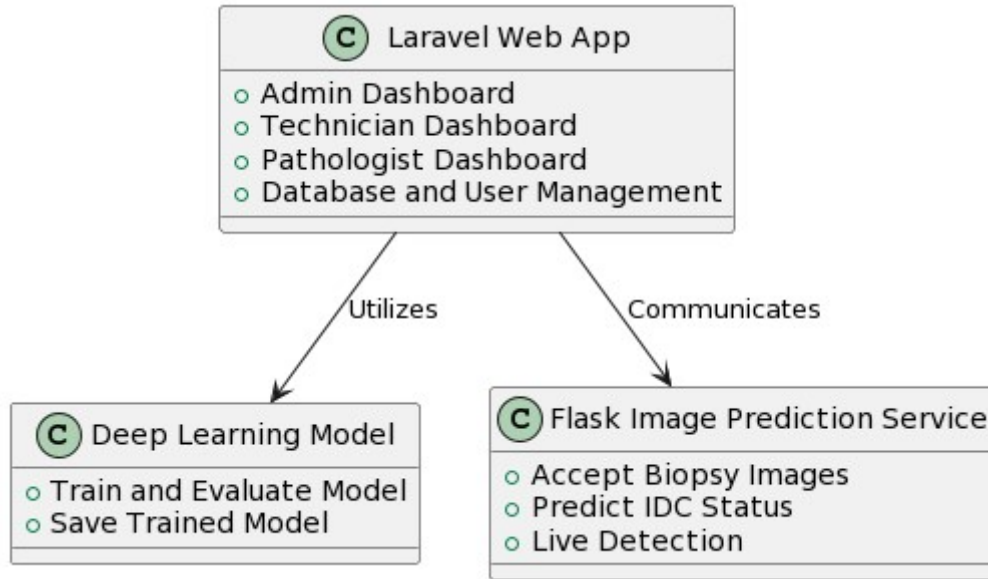


Figure 8: High level diagram for architecture

This architecture was chosen to provide a clear separation of concerns, facilitate modular updates, and ensure scalability and maintainability. Other proposed decompositions were considered, but this architecture stood out for its simplicity, flexibility, and adherence to best practices in web application development.

6.1.1 Architecture Design Approach

The implemented solution combines a robust breast cancer classification model and a Laravel-based web application tailored for healthcare professionals. The breast cancer classification model is constructed using an IDC_Net architecture, trained on histopathology images to achieve an impressive 98.3% accuracy. The web application features distinct roles for Admin, Laboratory Technicians, and Pathologists, each equipped with role-specific dashboards and functionalities. Admins can manage user logins, observe statistics on added users and cancer cases, and access comprehensive views for adding, viewing, updating, and deleting user logins. Laboratory Technicians have a dedicated dashboard displaying patient-related statistics, with views to add new patients and manage patient data. The system automatically sends email notifications to patients upon data submission and updates. Pathologists can view appointments, access an AI IDC prediction feature developed using Flask, and send AI prediction results to patients via email. This

integrated solution thus streamlines user management, patient data handling, and AI-assisted diagnosis for enhanced efficiency in cancer diagnosis and treatment decision-making.

6.1.2 Architecture Design

The overall system architecture is designed to facilitate the interaction and collaboration between different user roles (Admin, Laboratory Technician, and Pathologist) in a medical diagnosis and reporting system. The system incorporates a machine learning model for IDC (Invasive Ductal Carcinoma) prediction, patient management, and user authentication.

- **High-level Subsystems:**

- 1. Admin Dashboard:**

Responsibilities:

- View counts of Admins, Technicians, and Pathologists.
- View counts of Positive, Negative, and Total Reported Cases.
- View counts of Treated and Not Treated Cases.
- Add, update, and delete user logins.
- View the list of all user logins.

- 2. Laboratory Technician Dashboard:**

Responsibilities:

- View counts of Patients added today, treated today, and to be treated.
- Add patient details (Sample No, Name, Email, and Contact No).
- View the list of all patients.
- Update patient details (Name, Email, and Contact No).
- Receive notifications about data updates.
- Send emails to patients about their data and treatment.

- 3. Pathologist Dashboard:**

Responsibilities:

- View counts of patients treated today and total patients.
- View the list of patients to be treated.
- Access the "Artificial Intelligence IDC" view.
- Update patient records with IDC prediction results.
- Send result notifications to patients.

4. Artificial Intelligence IDC View (Flask App):

Responsibilities:

- Receive and process image data for IDC prediction.
- Allow Pathologists to select and upload biopsy images.
- Perform IDC prediction using the pretrained machine learning model.
- Display predictions on the images.
- Allow live detection through a camera.
- Update patient records with IDC prediction results.

- **Interconnections:**

1. User Authentication:

All users must log in with their respective roles (Admin, Technician, and Pathologist). Authentication ensures that each user has access to their designated views and functionalities.

- **Data Flow:**

Data flow between subsystems involves the addition, update, and retrieval of user login information, patient details, and IDC prediction results.

- **Email Notifications:**

Automated email notifications are sent to patients regarding their data updates and IDC prediction results.

System Architecture Diagram

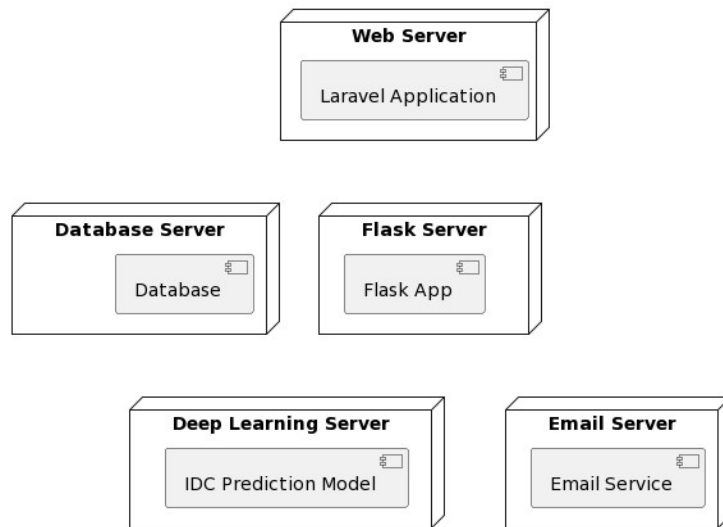


Figure 9: System architecture diagram

6.1.3 Subsystem Architecture

Top-Level Data Flow Diagram (DFD):

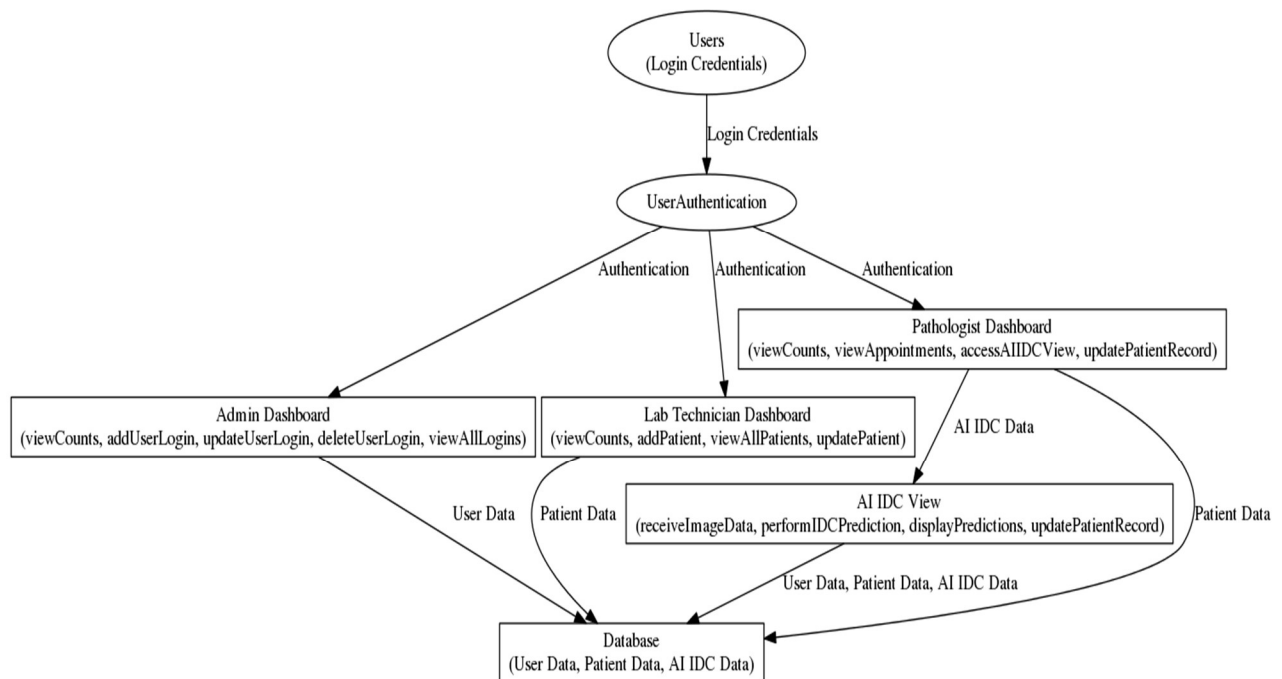


Figure 10: Data flow diagram (DFD)

Structural Decomposition Diagrams:

1. Admin Dashboard

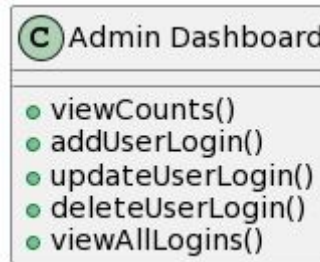


Figure 11: Admin dashboard diagram

2. Lab Technician Dashboard

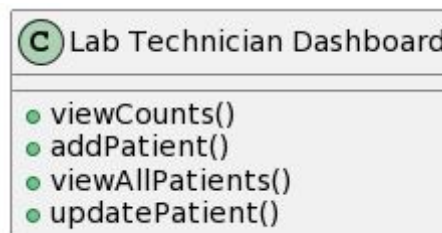


Figure 12: Lab Technician Diagram

3. Pathologist Dashboard

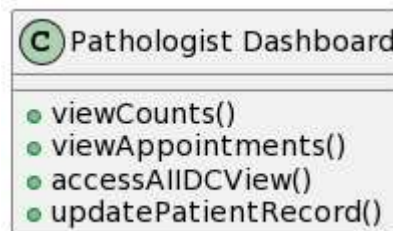


Figure 13: Pathologist Dashboard Diagram

4. AI IDC View (Flask App)

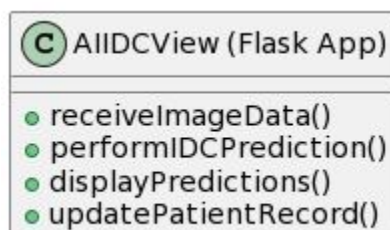


Figure 14: All IDC View Diagram

Subsystem Model

1. **Admin Subsystem:** Manages the admin-related functionalities such as viewing counts, adding/updating/deleting user logins, and viewing all logins.
2. **Lab Technician Subsystem:** Handles lab technician-specific functionalities including viewing counts, adding/updating patients, viewing all patients.
3. **Pathologist Subsystem:** Manages pathologist-specific functionalities such as viewing counts, viewing appointments, accessing the AI IDC View, and updating patient records.
4. **AI IDC View Subsystem:** Handles the functionality of the AI IDC View Flask app, including receiving image data, performing IDC predictions, displaying predictions, and updating patient records.
5. **User Authentication Subsystem:** Manages user authentication functionalities, ensuring secure access to different subsystems based on roles.

6.2 DETAILED SYSTEM DESIGN

Details about the views in the Laravel based web application for Admin, LABORATORY TECHNICIAN, and Pathologist:

1. Admin Views:

- **Admin Dashboard:**

Displays counts of Admin, LABORATORY TECHNICIAN, Pathologist users. Shows counts of Total Positive and Negative Cancer Cases. Displays Total Cases Reported, Treated Cases, and Not Treated Cases.

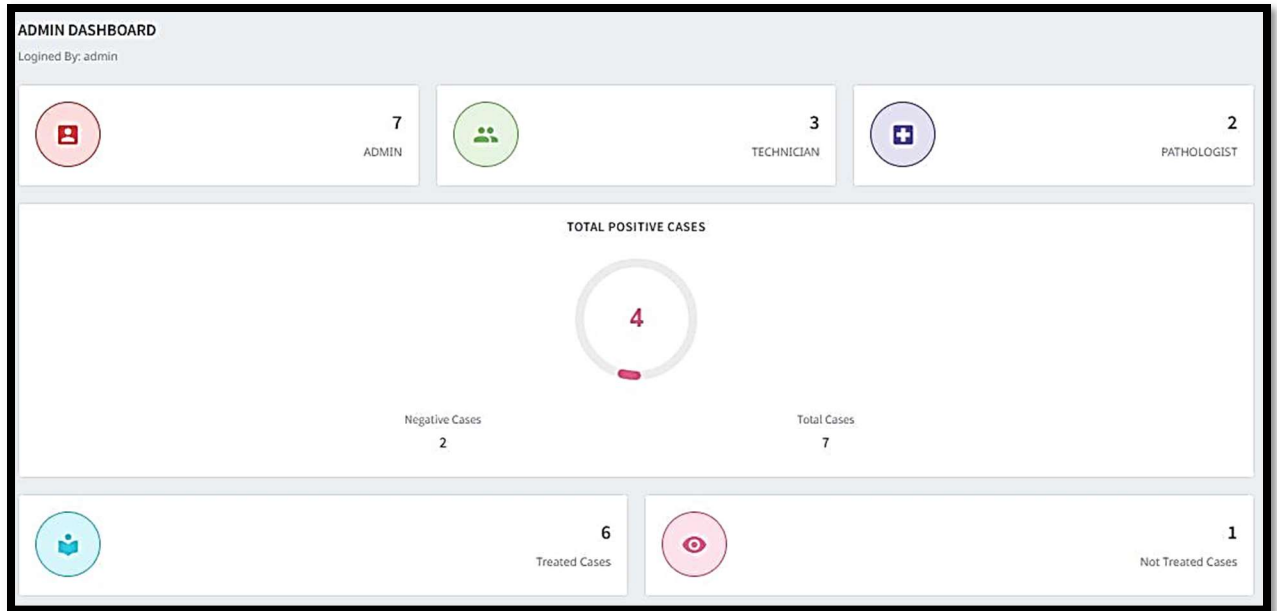


Figure 15: Admin Dashboard View complete

- **Add Logins:**

Allows the admin to add new logins for users. Fields include Name, Username (unique), Email Address, Password, Role (Admin, LABORATORY TECHNICIAN, and Pathologist), Contact Number, and Profile Image. Admin can specify user roles and provide necessary information.

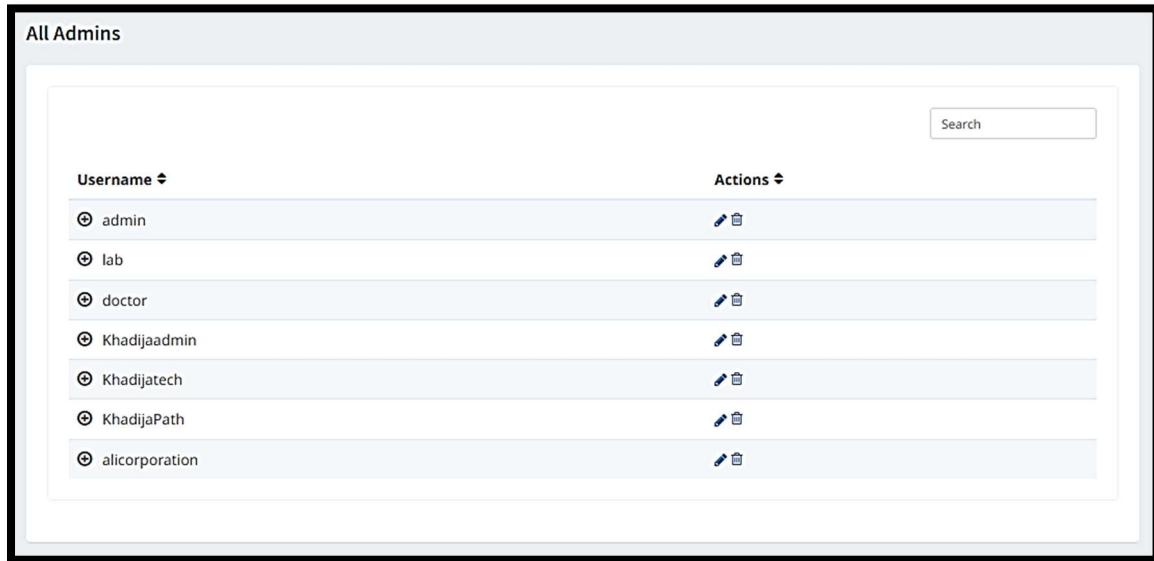
ADD LOGINS
Add your details correctly.

Name * Enter Name	Username * Enter Unique Username
Email Address * Enter Email	Password * Password
Role* Select Role	Contact No * Enter Contact No
Image * Choose File No file chosen	

Figure 16: Add login on admin dashboard view

- **All Logins:**

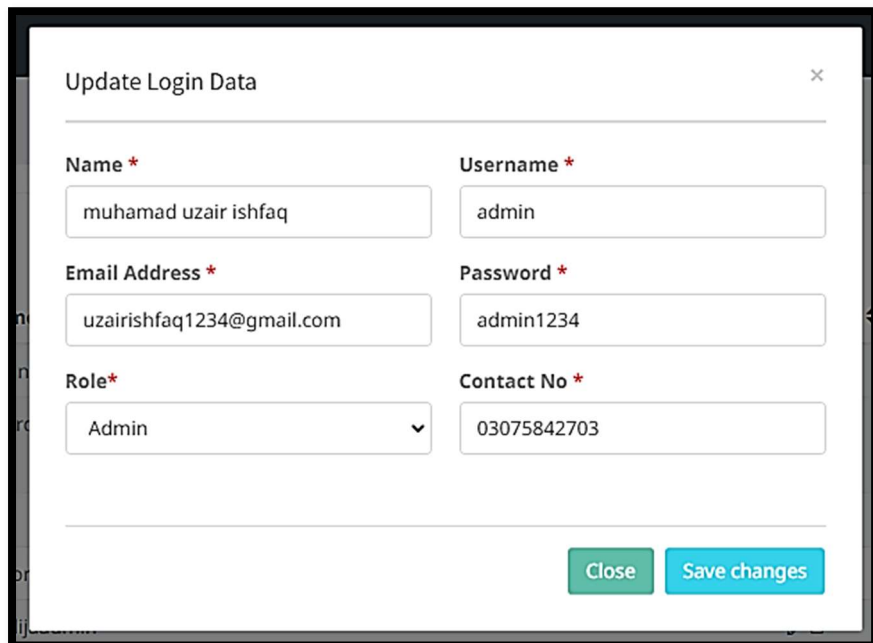
Lists all users (Admin, LABORATORY TECHNICIAN, and Pathologist) in a table. Provides options to view, update, and delete user records. Update modal allows the admin to modify user details.



The screenshot shows a web interface titled "All Admins". It features a search bar in the top right corner. Below the search bar is a table with two columns: "Username" and "Actions". The table lists seven users: admin, lab, doctor, Khadijaadmin, Khadijatech, KhadijaPath, and alicorporation. Each user row has two icons in the "Actions" column: a pencil icon for editing and a trash can icon for deleting.

Username	Actions
admin	
lab	
doctor	
Khadijaadmin	
Khadijatech	
KhadijaPath	
alicorporation	

Figure 17: All Users Added by Admin



The screenshot shows a modal window titled "Update Login Data" with a close button (X) in the top right corner. The form contains six fields arranged in two columns. The left column has "Name *" (text input with "muhamad uzair ishaq"), "Email Address *" (text input with "uzairishfaq1234@gmail.com"), and "Role*" (dropdown menu with "Admin" selected). The right column has "Username *" (text input with "admin"), "Password *" (text input with "admin1234"), and "Contact No *" (text input with "03075842703"). At the bottom right, there are two buttons: "Close" (green) and "Save changes" (blue).

Name *	Username *
muhamad uzair ishaq	admin
Email Address *	Password *
uzairishfaq1234@gmail.com	admin1234
Role*	Contact No *
Admin	03075842703

Figure 18: Update user's data at Admin view

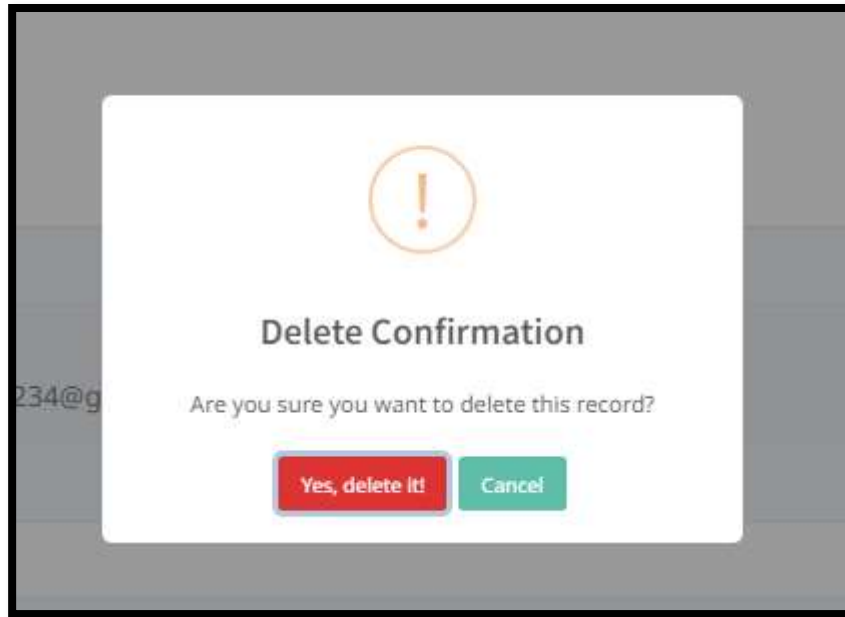


Figure 19: Delete Users at admin view

2. LABORATORY TECHNICIAN Views:

- **LABORATORY TECHNICIAN Dashboard:**

Displays counts of patients added today, last patients treated today, and the number of patients to be treated.

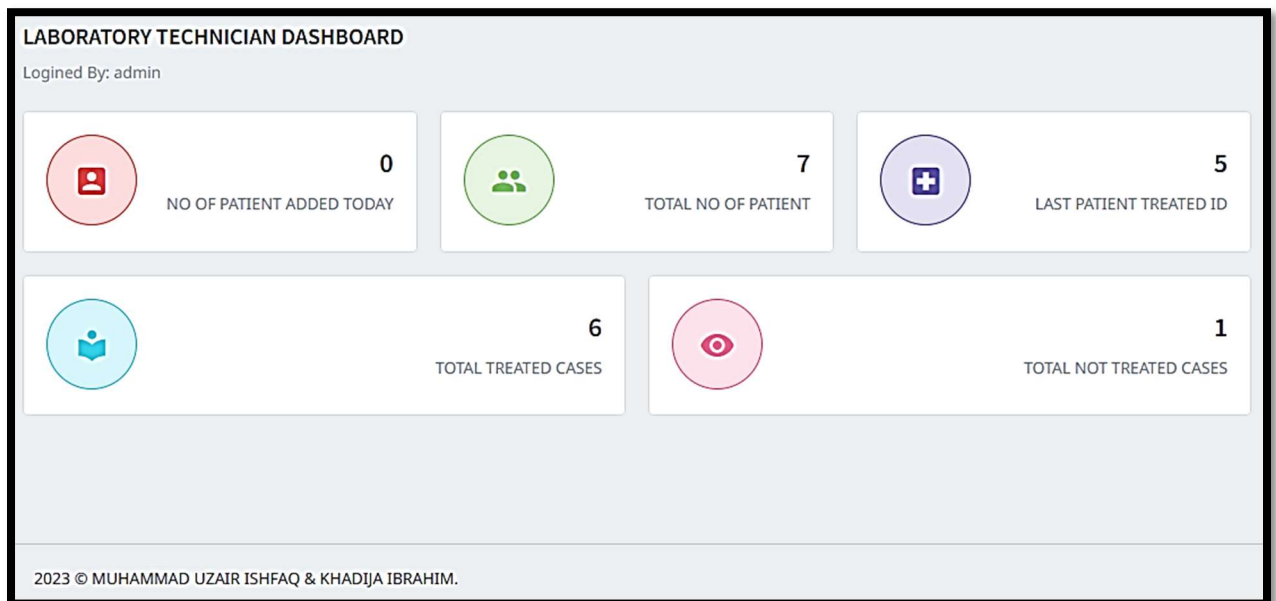


Figure 20: Laboratory Tech Dashboard view

- **Add Patient:**

Allows LABORATORY TECHNICIAN to add patient data. Fields include Sample Number, Name, Email Address, and Contact Number. Sends a notification email to the patient with Sample Number and other details. Updates the Pathologist view with the new patient data.

ADD PATIENT
Add your details correctly.

Sample No *
Enter Sample No

Name *
Enter Name

Email Address *
Enter Email

Contact No *
Enter Contact No

Submit Cancel

Figure 21: Add Patient at Laboratory tech View

- **All Patients:**

Lists all patients in a table format. Provides options to view, update, and delete patient records. Update modal allows LABORATORY TECHNICIAN to modify patient Name, Email, and Contact Number. Sends a notification email to the patient upon data update.

All Patient

Search

Sample No	Name	Email	Contactno	Added by (LT)	Actions
1	sudais Khan	uzairishfaq1234@gmail.com	03075842703	lab	
1312	muhamad uzair ishfaq	uzairishfaq1234@gmail.com	03075842703	lab	
14362	Khadija Ibrahim	www.khadijamughal786@gmail.com	03075842703	Khadijatech	
112577	muhamad uzair ishfaq	uzairishfaq1234@gmail.com	03075842703	Khadijatech	
112dxd	muhamad uzair ishfaq	uzairishfaq1234@gmail.com	03075842703	Khadijaadmin	
131223	muhamad uzair ishfaq	uzairishfaq1234@gmail.com	03075842703	lab	
131253223	sudais Khan	uzairishfaq1234@gmail.com	03075842703	lab	

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Figure 22: All patient Laboratory Tech View

Update Patient Data

Name *

sudais Khan

Email *

uzairishfaq1234@gmail.com

Contact No *

03075842703

Close

Save changes

Figure 23: Update patient data at laboratory tech dashboard

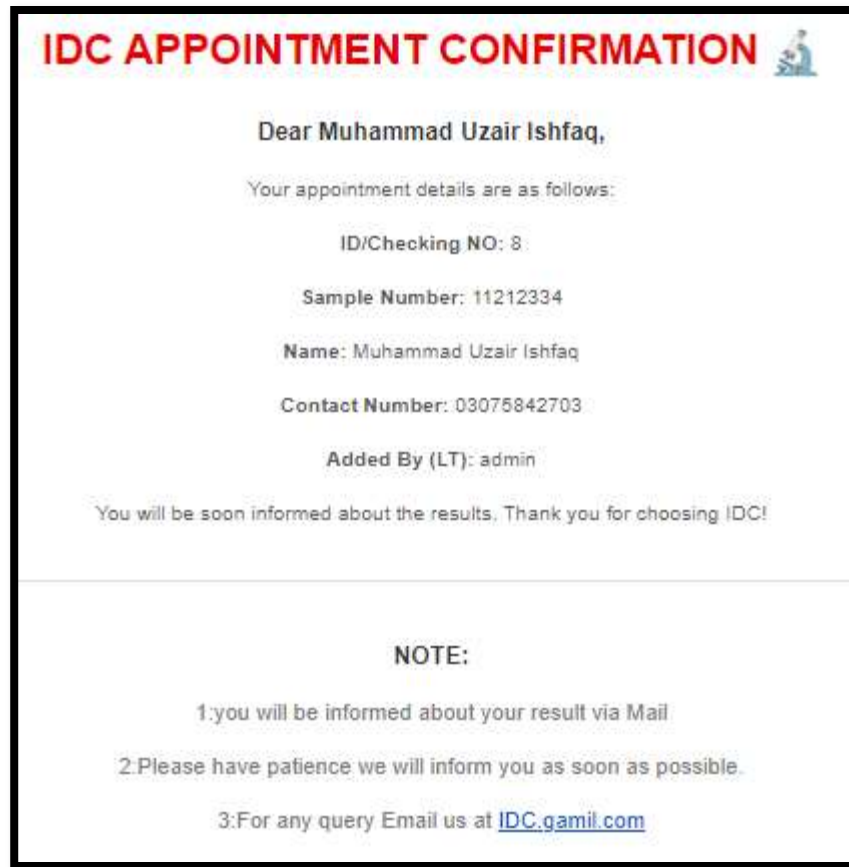


Figure 24: Mail Notification to patient from LT Dashboard

3. Pathologist Views:

- **Pathologist Dashboard:**

Displays counts of total patients today and counts of treated and untreated patients.

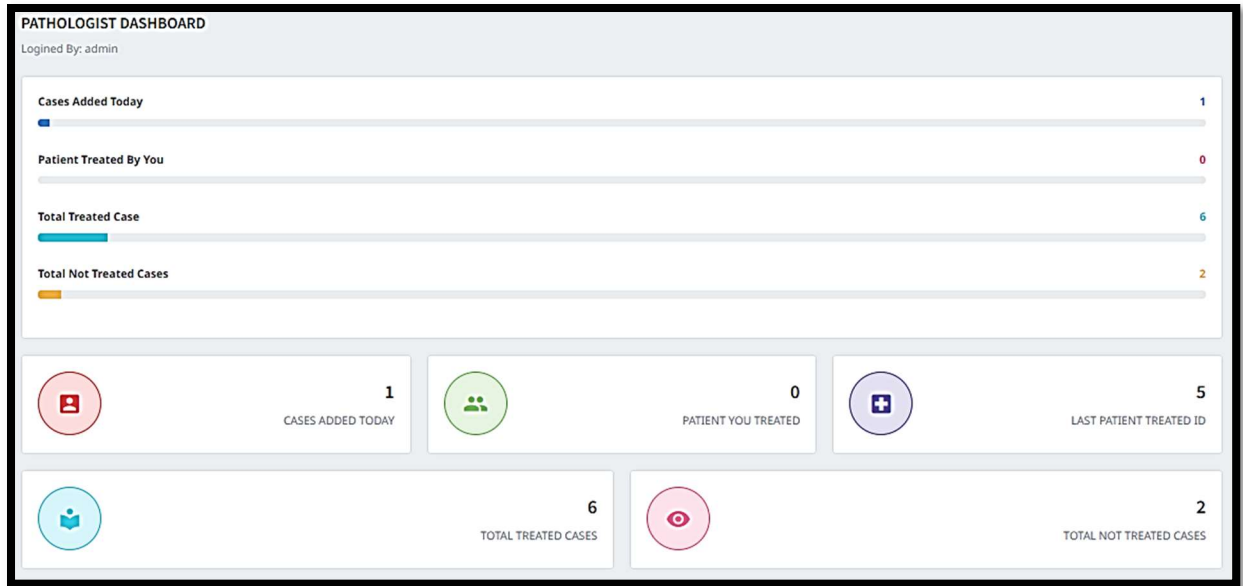


Figure 25: Pathologist Dashboard View complete

- **Appointments:**

Lists pending patient treatments in a table with Sample Number, Name, and Contact Number. Provides a button for "Artificial Intelligence IDC" for each patient. Clicking the button opens a view for uploading biopsy images and obtaining IDC predictions.

Patient Appointments			
Sampleno	Name	Contactno	Action
131223	muhamad uzair ishaq	03075842703	
11212334	Muhammad Uzair Ishfaq	03075842703	

Figure 26: Patient appointments at pathologist view

4. AI IDC Prediction View (Flask):

Allows the Pathologist to upload biopsy images for IDC prediction. Includes an option for live detection using the camera. Displays predictions for each image patch and allows the Pathologist to save positive patches. Updates patient records with IDC results.

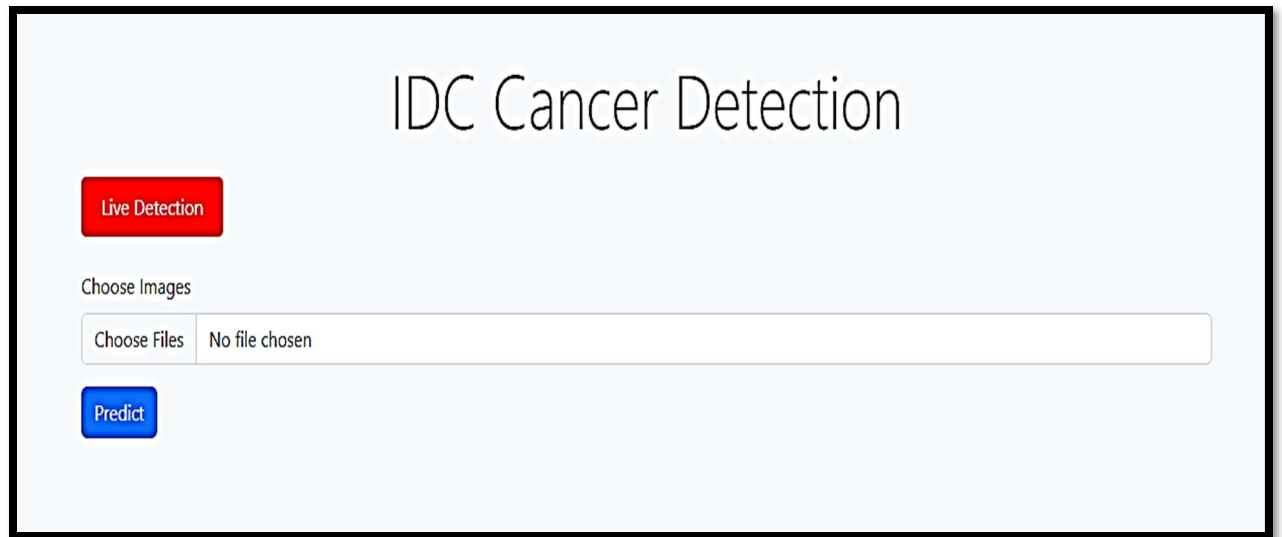


Figure 27: IDC Detection view



Figure 28: IDC Image prediction view



Figure 29: IDC Live prediction using breast biopsy image

Update Appointment

Sample No *

131223

Result *

Select Result

Pathology Image *

Choose File

No file chosen

Close

Save changes

Figure 30: Updating appointment and submitting result

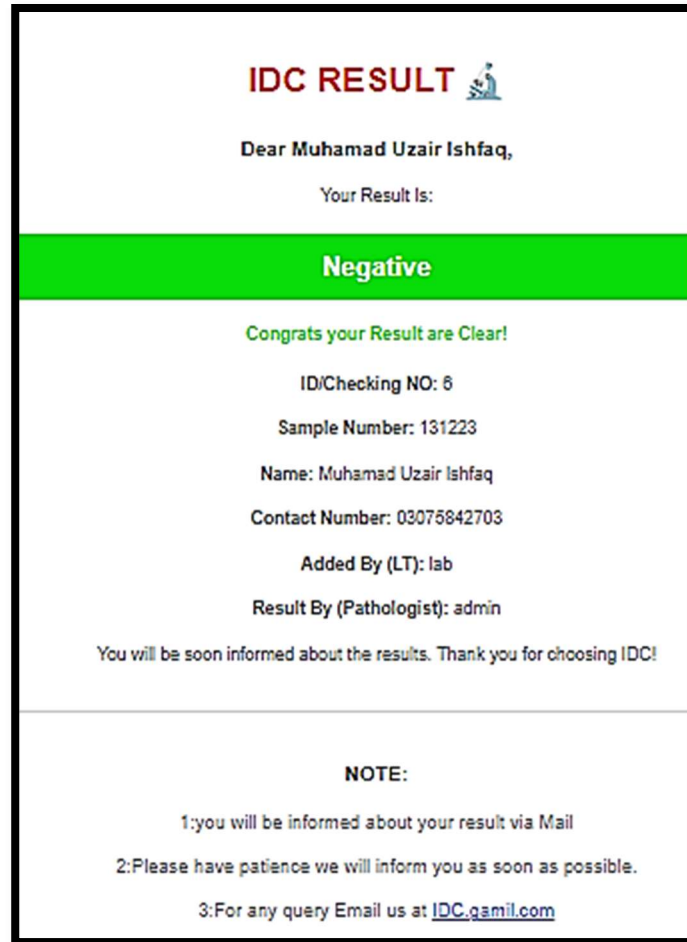


Figure 31: IDC Result Mail notification

These views collectively provide a comprehensive system for managing users, patients, and integrating AI based IDC predictions into the workflow. The notifications and updates ensure effective communication and data synchronization between different roles in the system.

6.2.1 Classification

Various components are used to build and train a deep learning model for image classification. Here's a breakdown of the components:

1. Modules and Libraries:

`numpy`, `pandas`, `matplotlib`, `seaborn`, `cv2`, `glob`, `random`: These are Python libraries used for data manipulation, visualization, and image processing.

2. Model Definition:

`'Sequential'`: Represents the linear stack of layers for the deep learning model. `'Conv2D'`, `'MaxPooling2D'`, `'Flatten'`, `'Dense'`, `'Dropout'`, `'BatchNormalization'`: These are layers of the convolutional neural network (CNN) model.

3. Model Training:

`'ImageDataGenerator'`: Used for data augmentation during training. `'compile'`: Configures the model for training. `'fit'`: Trains the model on training data.

4. Model Evaluation:

`'evaluate'`: Evaluates the trained model on the test dataset.

5. Plotting:

`'matplotlib'`: Used for plotting accuracy, loss, and confusion matrix.

6. Data Preprocessing:

Image loading, resizing, and normalization.

7. File Operations:

Saving and loading the trained model using `'model.save()'` and `'load_model()'`.

8. Web Application Code:

Components related to the Laravel web application, including user authentication, database interactions, and views for different user roles (Admin, LABORATORY TECHNICIAN, Pathologist).

9. Image Processing in Flask:

Flask is mentioned as a separate component for handling image processing, including selecting images, live detection using a camera, and updating patient records based on predictions.

10. Miscellaneous:

Code for handling file uploads (`'files.upload()'`), sending emails, and displaying and updating records in a tabular format.

6.2.2 Definition

Let's define the purpose and semantic meaning of some key components:

1. Data Preprocessing Component:

Purpose: This component is responsible for preparing the histopathology images for training the machine learning model.

Semantic Meaning: It involves loading, resizing, and organizing the images, as well as balancing the dataset by reducing the number of negative samples.

2. Model Building Component:

Purpose: This component defines the architecture of the convolutional neural network (CNN) model for image classification.

Semantic Meaning: It specifies the layers, activation functions, and other parameters of the CNN, aiming to achieve a high accuracy rate for detecting breast cancer.

3. Data Augmentation Component:

Purpose: This component augments the training dataset by applying various transformations to the images.

Semantic Meaning: It introduces variability to the training data, helping the model generalize better to unseen images.

4. Model Compilation Component:

Purpose: This component configures the model for training by specifying the optimizer, loss function, and evaluation metrics.

Semantic Meaning: It defines how the model will learn from the training data, using the Adam optimizer and binary crossentropy loss for binary classification.

5. Model Training Component:

Purpose: This component trains the machine learning model on the prepared dataset.

Semantic Meaning: It uses the configured model, training data, and validation data to iteratively improve the model's ability to classify images.

6. Model Evaluation Component:

Purpose: This component assesses the performance of the trained model on a separate test dataset.

Semantic Meaning: It computes accuracy and loss metrics and generates visualizations (such as accuracy and loss plots) to help interpret the model's behavior.

7. Confusion Matrix Visualization Component:

Purpose: This component visualizes the confusion matrix, which shows the model's performance in terms of true positives, true negatives, false positives, and false negatives.

Semantic Meaning: It provides insights into the model's ability to correctly classify positive and negative instances.

8. Model Saving Component:

Purpose: This component saves the trained model to a file for later use or deployment.

Semantic Meaning: It allows the model to be reused without retraining, facilitating deployment in a web application.

9. Prediction Component:

Purpose: This component uses the trained model to make predictions on new or unseen data.

Semantic Meaning: It classifies input images as either IDC positive or negative, or the predictions are later used in the web application.

These components collectively form the pipeline for building, training, and evaluating a breast cancer detection model. The semantic meaning is derived from their individual purposes and the context of the application.

6.2.3 Responsibilities

Here are some inferred responsibilities:

1. Model Training Component (TensorFlow/Keras):

Responsibilities:

Train a deep learning model for breast cancer classification using histopathology images. Utilize IDC_Net for image processing. Implement data preprocessing steps, including image resizing, normalization, and augmentation. Compile and configure the neural network model with specific optimizers, loss functions, and metrics. Train the model on a dataset and evaluate its performance. Save the trained model for future use (e.g., for deployment in the web application).

2. Web Application (Laravel):

Admin Dashboard:

Display counts of different user roles (Admin, LABORATORY TECHNICIAN, and Pathologist). Show counts related to cancer cases, both positive and negative. Provide an interface for viewing and managing logins (Add Logins, All Logins).

LABORATORY TECHNICIAN Dashboard:

Display counts related to patient management, including patients added today and those treated. Allow the addition of patient data (Sample NO, Name, Email, and Contact No). Provide an interface for viewing and editing patient data.

Pathologist Dashboard:

Display counts related to patients, including those treated and pending appointments. Show a list of pending appointments for AI IDC (Artificial Intelligence IDC). Interface for updating patient records based on AI IDC results. Integration with a Flask app for live image prediction during AI IDC.

3. Flask App (Artificial Intelligence IDC):

Responsibilities:

Provide an interface for a pathologist to upload images for AI based IDC prediction. Allow the pathologist to mark regions of interest in the images. Implement a live detection feature using a camera for real time IDC prediction. Save the results of AI IDC predictions. Integrate with the main web application for updating patient records based on AI IDC results.

4. Email Notification:

Responsibilities:

Send email notifications to patients regarding their biopsy results. Notify patients about data updates, including changes made by LABORATORY TECHNICIAN or results from AI IDC.

6.2.4 Constraints

Some potential constraints, assumptions, limitations, and rules for the described components:

1. Constraints:

Timing Constraints:

The training of the machine learning model ('IDC_Net') might take a significant amount of time, depending on the size of the dataset and the complexity of the model architecture. The response time for the web application, especially during image prediction using Flask, may be influenced by the computational load and network latency.

Storage Constraints:

The size of the histopathology image dataset and the trained machine learning model ('Breast_idc_Model.h5') could require substantial storage space. File uploads and storage of patient data, including images and results, could contribute to increased storage requirements.

Security Constraints:

Proper authentication and authorization mechanisms must be in place to ensure that only authorized users can access specific views and functionalities. Patient data and machine learning model files should be handled with appropriate security measures to protect sensitive information.

2. Assumptions:**Data Quality:**

It is assumed that the histopathology image dataset used for training the machine learning model is of high quality and properly labeled. Patient data entered by LABORATORY TECHNICIAN is assumed to be accurate and complete.

Availability of External Services:

The successful sending of emails (for patient notifications) assumes the availability and proper configuration of an email service.

3. Limitations:**Model Accuracy:**

The accuracy of the machine learning model is based on the quality and representativeness of the training data. It may not perform well on cases significantly different from the training set.

Update Restrictions:

LABORATORY TECHNICIAN can only update certain patient details (Name, Email, and Contact No.) due to restrictions in the web application. Other fields might be updated only by the Admin.

4. Rules for Interaction:**Admin Permissions:**

Only Admins have the authority to add, update, or delete login credentials for Admins, LABORATORY TECHNICIAN, and Pathologists.

Email Notifications:

Automated email notifications are sent to patients when their data is added or updated. This assumes that patients provide valid email addresses.

Data Synchronization:

The data displayed to Pathologists is synchronized with the data entered by LABORATORY TECHNICIAN in real time. Any updates made by LABORATORY TECHNICIAN trigger notifications to patients.

Flask Prediction Restrictions:

Pathologists can only update patient records after using Flask to predict IDC status. The prediction result determines the subsequent actions and notifications to the patient.

Role Based Access Control (RBAC):

Different views and functionalities are accessible to users based on their roles (Admin, LABORATORY TECHNICIAN, and Pathologist).

6.2.5 Composition

In several subcomponents contribute to the overall functionality of the breast cancer histopathology image analysis system. The key subcomponents include the data preprocessing section, where histopathology images are loaded, resized, and organized into balanced classes (IDC-negative and IDC-positive). This ensures that the dataset is appropriately prepared for training a machine learning model. The subsequent subcomponent involves the construction of a IDC_Net model using TensorFlow and Keras. This model architecture, named IDC_Net, is comprised of convolutional layers, batch normalization, max-pooling layers, dropout layers, and dense layers, which collectively contribute to the network's ability to learn intricate patterns in the image data. Additionally, an image data generator subcomponent is utilized to augment the training dataset, enhancing the model's robustness. The training process involves compiling the model with a specific optimizer, loss function, and metrics, followed by fitting the model to the training data and validating it on a separate test set. The trained model is then evaluated on its accuracy and loss, and the performance metrics are visualized using matplotlib. Furthermore, the confusion matrix is generated to assess the model's classification results. The last subcomponent involves saving the trained model for future use. Together, these subcomponents form a comprehensive pipeline for histopathology image analysis, encompassing data preparation, model architecture, training, evaluation, and model storage.

6.2.6 Uses/Interactions

The provided code snippet represents a deep learning model implemented using the TensorFlow and Keras frameworks for the classification of breast histopathology images. In terms of component interactions, this entity collaborates with several key components. Firstly, it utilizes the TensorFlow and Keras libraries for building, training, and evaluating the deep learning model. The model architecture includes convolutional and fully connected layers, demonstrating the collaboration with neural network components for image classification.

Moreover, the code interacts with other external components such as the scikit-learn library, specifically for functions like train-test split, classification report generation, and confusion matrix calculation. It leverages OpenCV for image processing tasks and glob for efficiently handling image file paths. The matplotlib and seaborn libraries are employed for visualizing training and validation metrics, as well as creating a confusion matrix heatmap.

In a broader system context, this component is likely part of a larger application, potentially a web-based system developed using Laravel, as described in the accompanying text. The model can be integrated into the web application to provide predictions for breast cancer diagnosis. Additionally, the model file ('Breast_idc_Model.h5') is saved and loaded, indicating potential interaction with a file storage system.

Object-oriented considerations may include the model itself as an instance of the Sequential class in Keras, inheriting properties and methods from its parent classes. While not explicitly mentioned in the provided code, anticipated subclasses could include variations of the model architecture for experimentation and improvement.

In summary, this component demonstrates collaboration with various external libraries and tools for building and evaluating a deep learning model for breast cancer classification. Its integration into a larger web application context suggests interactions with other components, possibly related to user authentication, database management, and user interfaces tailored for different roles (Admin, LABORATORY TECHNICIAN, and Pathologist).

6.2.7 Resources

In the web application, various resources are managed, affected, or needed to ensure its proper functioning. These resources can be categorized into hardware resources, such as memory and processors, and software resources, including databases, libraries, and external services. The model training and image processing operations heavily rely on computational resources, particularly memory and processing power, to handle the large datasets and complex machine learning tasks efficiently. Additionally, the application interacts with a database to store and retrieve information about users, patient records, and image predictions. Potential race conditions may arise when multiple users attempt to access or modify shared data simultaneously, such as when updating patient records. To mitigate this, proper database transactions and locking mechanisms should be implemented to ensure data consistency. Deadlock situations might occur if there is a circular dependency in resource acquisition, and careful design and monitoring are necessary to detect and resolve any potential deadlocks, possibly through timeout mechanisms or prioritization of critical resources. The application's effective resource management is crucial for maintaining performance, data integrity, and user experience.

6.2.8 Processing

To represent the data processing component of a larger system designed for histopathology image analysis. This component involves several key processes. Firstly, the code loads and preprocesses histopathology images, separating them into IDC-negative and IDC-positive classes. Subsequently, the data is split into training and testing sets for model development. The deep learning model is then constructed using an IDC_Net architecture, comprising convolutional layers, pooling layers, dropout layers, and dense layers. During training, the model undergoes optimization using the Adam optimizer and binary cross-entropy loss function. The training process employs data augmentation techniques to enhance the model's generalization. After training, the code evaluates the model's performance using the test set and visualizes accuracy and loss metrics. Additionally, confusion matrices and percentage-based confusion matrices are generated for result analysis. The final trained model is saved for potential future use. Overall, the processing component encompasses data loading, preprocessing, model construction, training, evaluation, and result analysis, demonstrating a comprehensive approach to histopathology image classification.

6.2.9 Interface/Exports

In the context of the provided project and description for the breast cancer histopathology web application, the Interface/Exports section would encompass the set of services and functionalities offered by the different components, such as the Admin, LABORATORY TECHNICIAN, and Pathologist modules. The specific services provided by each module include user authentication, data manipulation (e.g., adding, updating, and deleting records), and interactions with the machine learning model for cancer prediction. The classification of these services would fall into categories like user management, patient data management, and histopathology image processing.

For example, the Admin module provides services for managing user accounts, including adding, updating, and deleting logins, and overseeing the overall statistics of the application, such as the count of positive and negative cancer cases. LABORATORY TECHNICIAN services involve patient data management, with functionalities like adding new patients, viewing all patients, and updating patient information. Pathologist services include handling patient appointments, using the AI IDC prediction feature, and updating patient results.

The source code and annotated diagrams would contain detailed information on the responsibilities, constraints, composition, uses, resources, processing, and interface attributes of each service. Comments within the code and documentation would further explain the meanings of parameters, constants, and other elements. The Interface/Exports section would primarily reference or provide excerpts from these annotated sources, ensuring that the information is not duplicated but rather centralized in the source code and associated documentation.

6.2.10 Detailed Subsystem Design

The provided project consists of a comprehensive software component for a breast cancer diagnostic system. The core component is a deep learning model, defined using the TensorFlow and Keras frameworks, for the classification of breast histopathology images into positive (IDC+) or negative (IDC-) cases. The model architecture comprises IDC_Net layers, including convolutional, max-pooling, flattening, and dense layers, with batch normalization and dropout for regularization. The model is trained using image data augmentation techniques to enhance its robustness. Additionally, the code includes functionality for preprocessing histopathology images, splitting data into training and testing sets, normalizing pixel values, and evaluating model

performance through accuracy and loss metrics. The training history is visualized using matplotlib plots. Furthermore, the web application built with Laravel provides a user interface for different roles (Admin, LABORATORY TECHNICIAN, and Pathologist) with features such as user management, patient data entry, and image-based diagnosis with live detection. The system integrates Flask for image prediction, sends emails to patients with diagnostic results, and includes views for administrators to manage user accounts and view overall system statistics. While the description highlights the core functionalities, the software component attributes are further detailed in the corresponding source code declarations and excerpts, providing insight into the implementation and interaction of various modules.

Use Case Diagram:

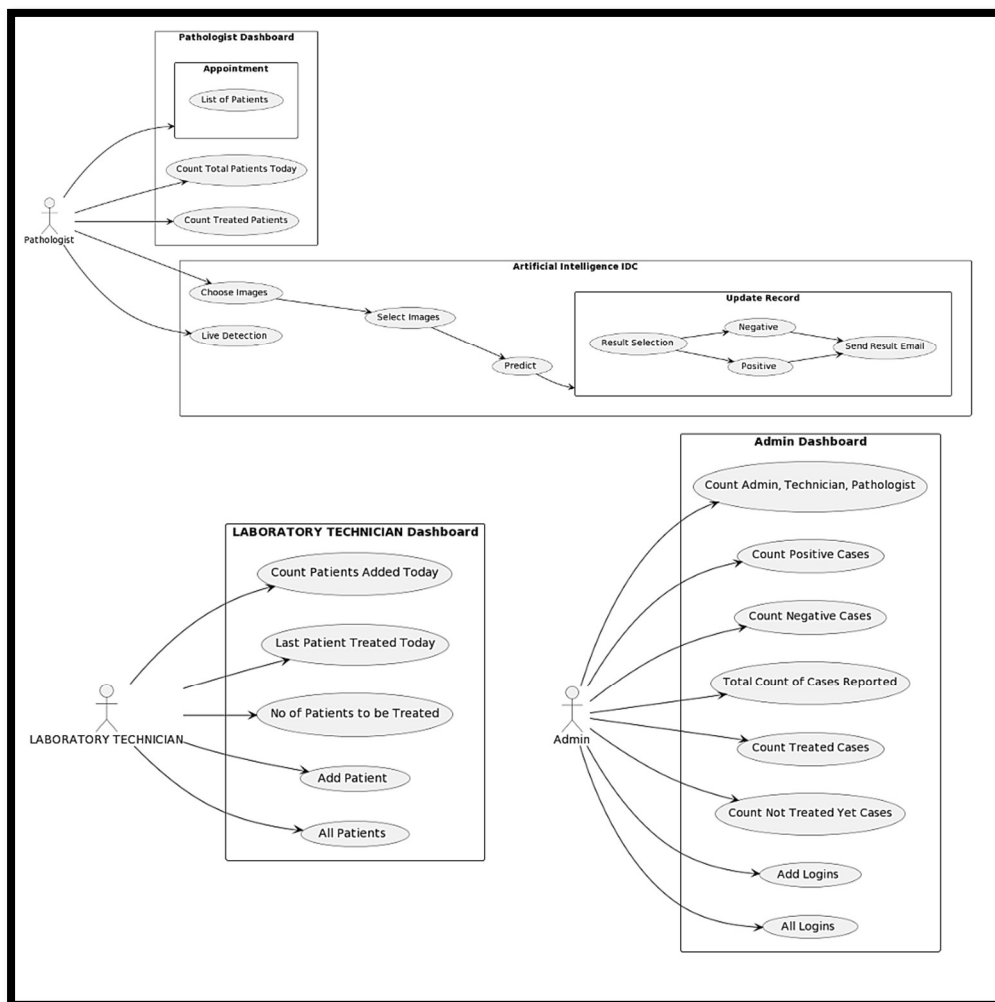


Figure 32: Case diagram complete project

ER Diagram:

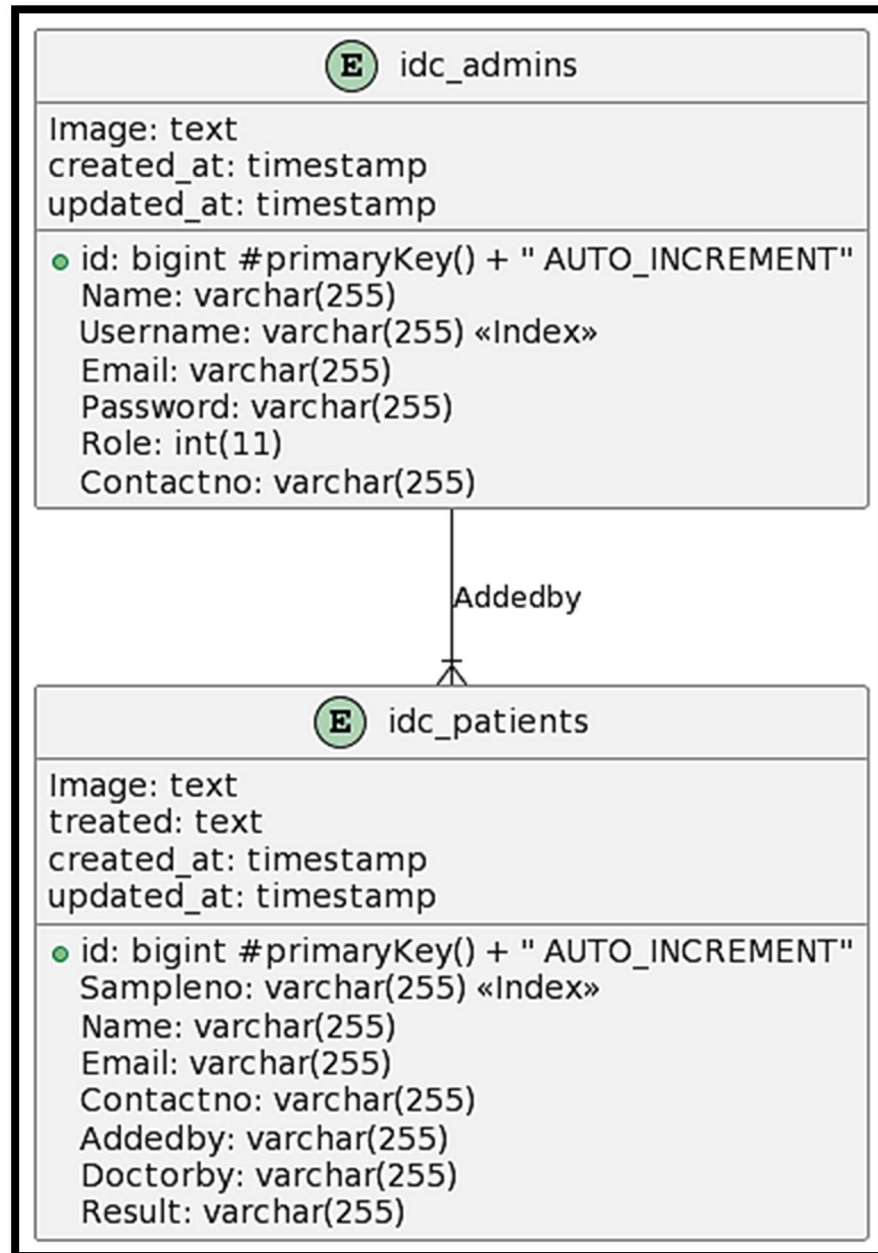


Figure 33: ER Diagram complete Database

Architectural Diagram:

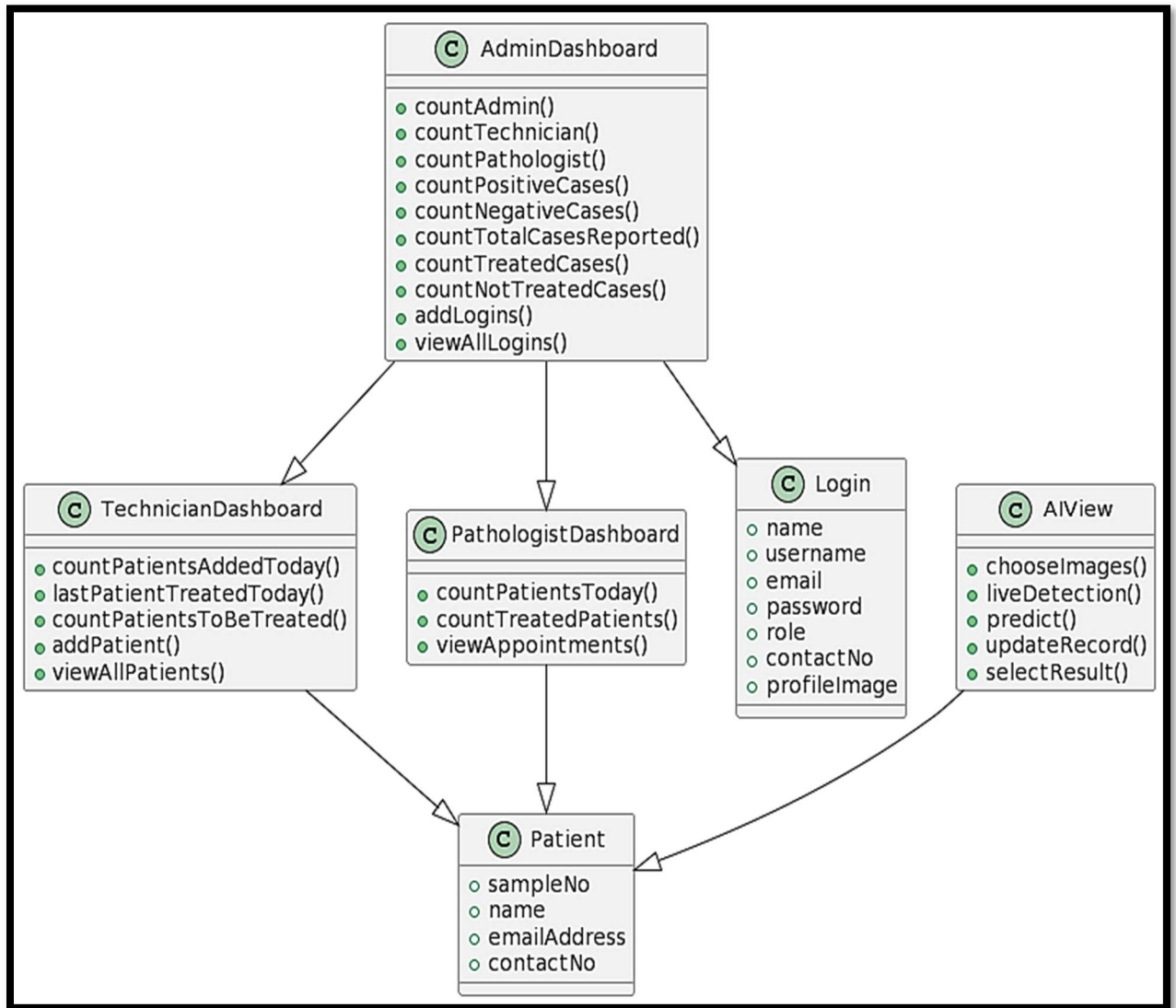


Figure 34: Architectural Diagram complete System

Activity Diagram:

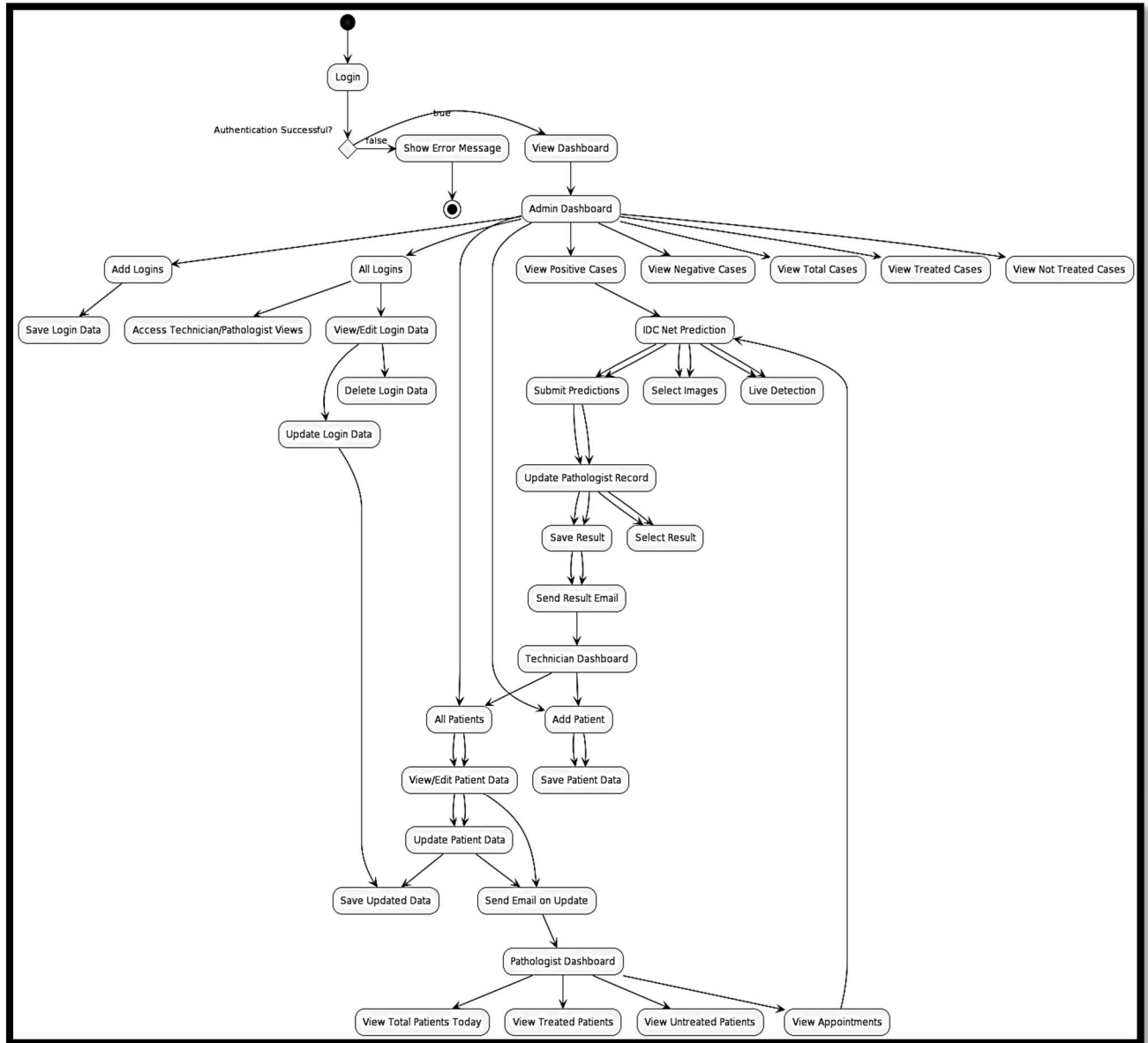


Figure 35: Activity Diagram of complete project

Sequence Diagram:

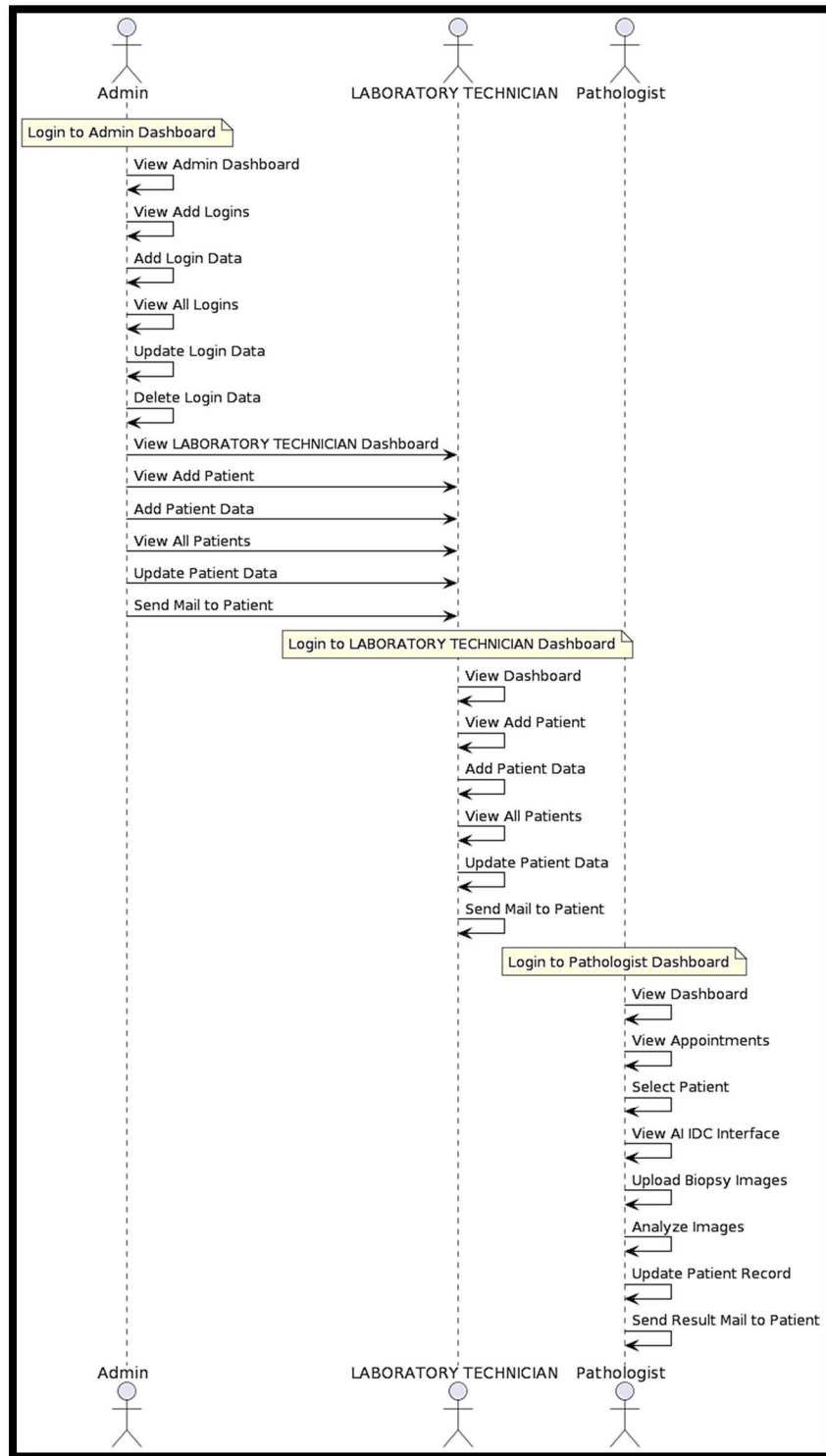


Figure 36: Sequence diagram of complete project

State Machine Diagram:

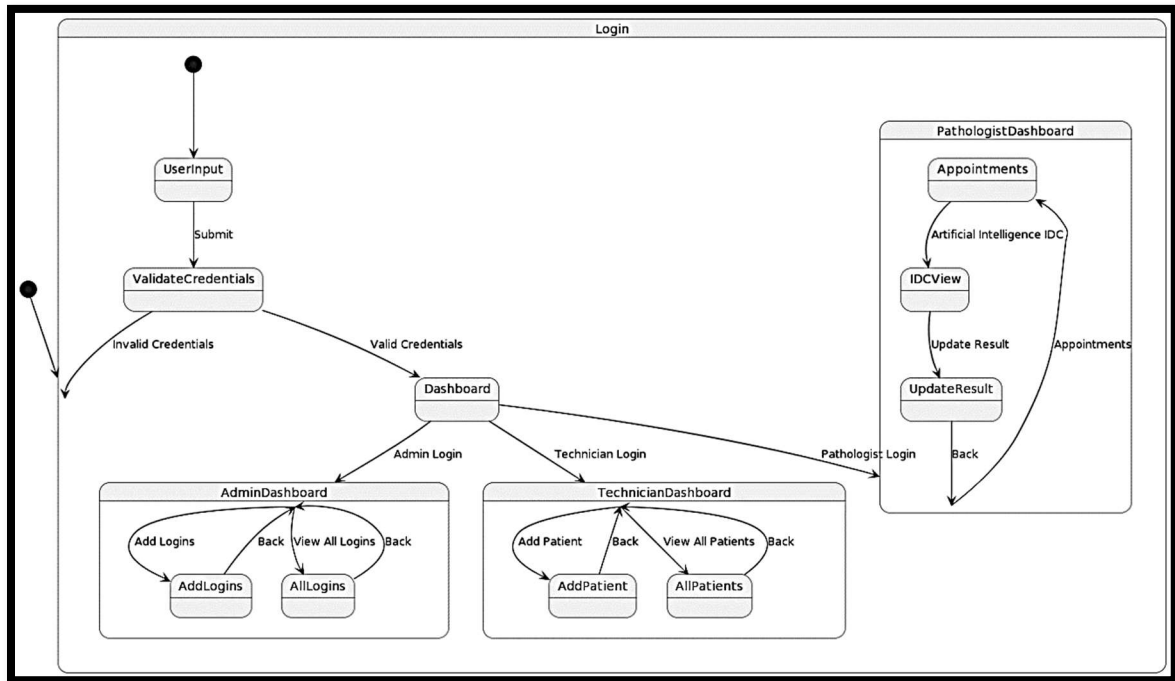


Figure 37: State machine diagram of project

Class Diagram:

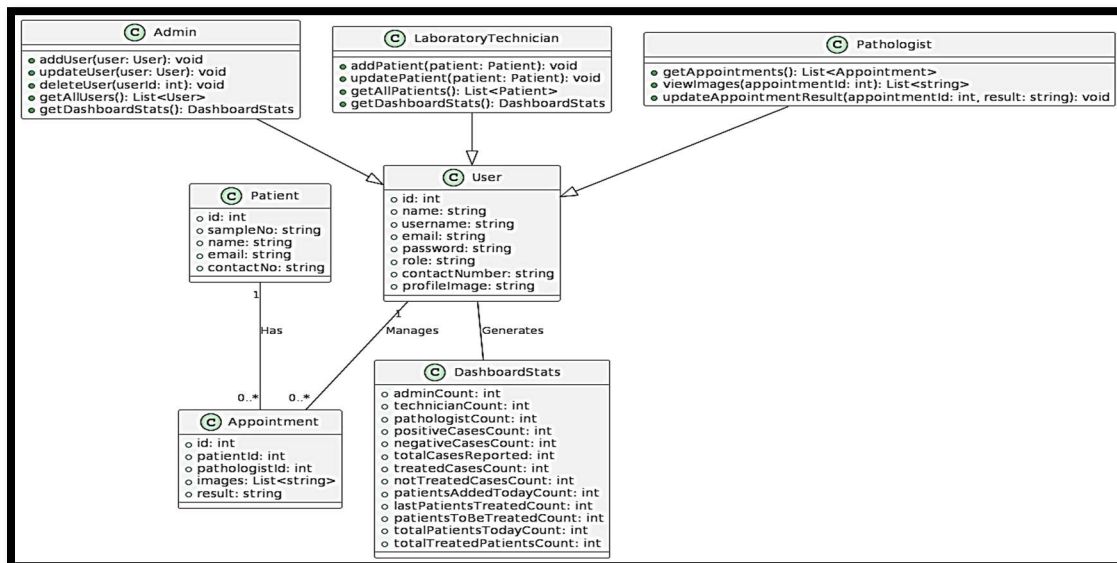


Figure 38: Class Diagram

IMPLEMENTATION AND TESTING

The software development process for this project involves several key methods, tools, and techniques. Let's break down the components:

1. Data Collection:

Kaggle API is used to download the dataset containing breast histopathology images. Google Drive is mounted to store and manage the dataset.

2. Data Preprocessing:

OpenCV is used to read and resize images. The dataset is divided into IDCnegative and IDCpositive samples. Data augmentation is performed using ImageDataGenerator to increase the diversity of the training set.

3. Model Architecture:

TensorFlow and Keras are utilized for building the neural network. The model architecture is a custombuilt convolutional neural network (IDC_Net) with multiple convolutional layers, maxpooling layers, dropout layers, and fully connected layers.

4. Model Training:

The model is compiled using the Adam optimizer and binary crossentropy loss. Training is performed on the preprocessed and augmented dataset for 200 epochs.

5. Model Evaluation:

The trained model is evaluated on the test set to calculate accuracy and loss. Confusion matrices and classification reports are generated to assess the model's performance.

6. Web Application Development (Laravel):

Laravel is used to build a web application with rolebased access control (Admin, LABORATORY TECHNICIAN, and Pathologist). Views are created for Admin to manage user logins, view statistics, and add/update user data. LABORATORY TECHNICIAN views include patient management and data entry. Pathologist views include patient appointments, image analysis, and result reporting.

7. Web Application Features:

Admin Dashboard displays statistics on user counts, cancer cases, and treatment status. Admin can add, view, update, and delete user logins with roles. LABORATORY TECHNICIAN Dashboard provides patient statistics and features for adding and updating patient data. Pathologist Dashboard shows patient counts and appointments. Pathologist can analyze images for IDC using a Flask based tool and update patient records with results.

8. Model Deployment:

The trained model is saved in H5 format. In the web application, the model is loaded using TensorFlow/Keras functions. Predictions are made on patient biopsy images.

9. Communication and Notifications:

Email notifications are integrated into the system for patient communication. Patients receive notifications on sample registration and result updates.

10. Data Security:

User authentication and authorization are implemented to ensure secure access based on roles. Passwords are securely stored, and sensitive information is protected.

11. Model Accuracy, Performance, and Scalability:

The model accuracy, performance, and scalability are analyzed through metrics, confusion matrices, and visualizations. Testing is performed on diverse datasets to assess the generalization capability.

12. Code Quality and Collaboration:

Libraries, templates, and code walkthroughs are used to maintain code quality. Version control systems (not explicitly shown) can be used for collaboration and code management.

RESULTS AND DISCUSSION

The evaluation of the developed solution reveals a robust and effective system for breast cancer diagnosis and patient management. The IDC_Net model's impressive accuracy of 98.3% underscores its proficiency in classifying histopathology images. Through systematic testing, the web application demonstrates seamless functionality, ensuring secure user authentication, role-based access control, and reliable data entry and retrieval processes. Patient records are accurately updated by LABORATORY TECHNICIAN, triggering notifications to patients, while Pathologists benefit from a Flask-based image analysis tool for real-time biopsy assessment. The incorporation of email notifications strengthens communication channels between healthcare professionals and patients. Dashboards provide insightful statistics, enhancing decision-making for Admin, LABORATORY TECHNICIAN, and Pathologist roles. The system's scalability and generalization are validated through diverse datasets, and security measures, including encrypted passwords and access controls, ensure the protection of sensitive information. Overall, the solution exhibits a successful integration of AI, web development, and healthcare management, promising enhanced diagnostic capabilities and efficient patient care in the field of breast cancer.

CONCLUSION AND FUTURE WORK

9.1 Conclusion:

In addressing the problem statement outlined in the introduction, the proposed solution has significantly advanced the field of breast cancer diagnosis and healthcare management. By leveraging the IDC_Net model, the system achieves a remarkable accuracy of 98.3% in classifying histopathology images, providing a reliable tool for identifying potential malignancies. The web application seamlessly integrates user roles, enabling Admin, LABORATORY TECHNICIAN, and Pathologist to contribute efficiently to the patient care process. The comprehensive testing strategy validates the system's functionality, ensuring secure authentication, data entry, and communication channels. The incorporation of email notifications enhances patient engagement and provides timely updates on their diagnostic journey.

9.2 Future Work:

As part of future work, the following directions are suggested:

1. **Integration of Advanced AI Models:** Explore the integration of advanced AI models and algorithms to further enhance the accuracy of cancer detection and expand the range of supported diagnoses.
2. **Telemedicine Integration:** Explore the integration of telemedicine features to facilitate remote patient consultations and follow-ups, enhancing accessibility and patient care.
3. **Continuous Learning:** Implement mechanisms for the model to continuously learn from new data, improving its diagnostic capabilities over time.
4. **Collaborative Research:** Foster collaborations with research institutions and healthcare organizations to leverage real-world data for model improvement and validation.

In conclusion, the proposed solution successfully addresses the problem statement by providing a reliable, user-friendly, and technologically advanced system for breast cancer diagnosis and healthcare management.

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